第四章 气体和蒸汽的基本的热力过程

Basic thermodynamic process

本章主要内容

4-1 理想气体的可逆多变过程

4-2 理想气体的定压、定容和定温过程

4-3 理想气体的定熵（可逆绝热）过程

4-4 理想气体热力过程综合分析

4-5 水蒸气的基本过程

4–1 理想气体的可逆多变过程

一、研究热力过程的目的和方法

1. 目的:

提高热力学过程的热功转换效率

热力学过程受外部条件影响

主要研究外部条件对热功转换的影响

利用外部条件，合理安排过程，形成最佳循环

对已确定的过程，进行热力计算

2. 对象：

① **参数**（p、T、V、U、H、S）**变化**

② **能量转换关系** q、w、w_t

3. 方法：

① **将热力过程模化为理想热力过程**

② **引入可逆假定进行分析计算**

③ **引入经验系数进行不可逆修正**

4. 研究步骤

①列出过程方程式：根据过程特点，一般整理成 p, v 的表达 式，即$f(p, v)=0$;

②建立基本状态参数间的关系式：过程方程 + 状态方程

③在p-v，T-s图上绘出过程曲线：可清晰地看出各过程状态变化及能量转换状况，即参数的变化趋势，q，w，w_t的正负。

④计算 Δu，Δh，Δs 的值：为q，w，w_t的计算打 基础。

⑤计算过程中能量的传递和转换量q，w，w_t：闭口系功的输出形式对应为w，开口系功的输出形式对应为w_t。

5. 可用的公式

①热力学第一定律

$$Tds \overset{\text{可逆}}{=\!=\!=} cdT = \delta q = du + \delta w \overset{\text{可逆}}{=\!=\!=} du + pdv$$

$$\overset{\text{理气}}{=\!=\!=} c_V dT + pdv$$

$$Tds \overset{\text{可逆}}{=\!=\!=} cdT = \delta q = dh + \delta w_t \overset{\text{可逆}}{=\!=\!=} dh - vdp$$

$$\overset{\text{理气}}{=\!=\!=} c_p dT - vdp$$

②理想气体状态参数的计算公式

$$
\begin{cases}
pv = R_g T \\
c_p - c_V = R_g \\
\Delta u = c_V \Delta T \qquad \Delta h = c_p \Delta T \\
\Delta s = c_p \ln \dfrac{T_2}{T_1} - R_g \ln \dfrac{p_2}{p_1} = c_V \ln \dfrac{T_2}{T_1} + R_g \ln \dfrac{v_2}{v_1} \\
\qquad\qquad\qquad\qquad = c_V \ln \dfrac{p_2}{p_1} + c_p \ln \dfrac{v_2}{v_1}
\end{cases}
$$

计算过程中能量的传递和转换量q，w，w_t：

计算q：

$$\begin{cases} q=\int c\mathrm{d}T=\begin{cases}\int c_p\mathrm{d}T & \text{定压}\\ \int c_V\mathrm{d}T & \text{定容}\end{cases} \\ q \overset{\text{可逆}}{=\!=\!=} \int T\mathrm{d}s \overset{\text{等温}}{=\!=\!=} T\Delta s \\ q=\Delta u+w \ =\Delta h+w_t \end{cases}$$

计算w, w_t：

$$w \overset{\text{可逆}}{=\!=\!=} \int p\mathrm{d}v=\begin{cases}p\Delta v & \text{定压}\\ 0 & \text{定容}\end{cases}$$

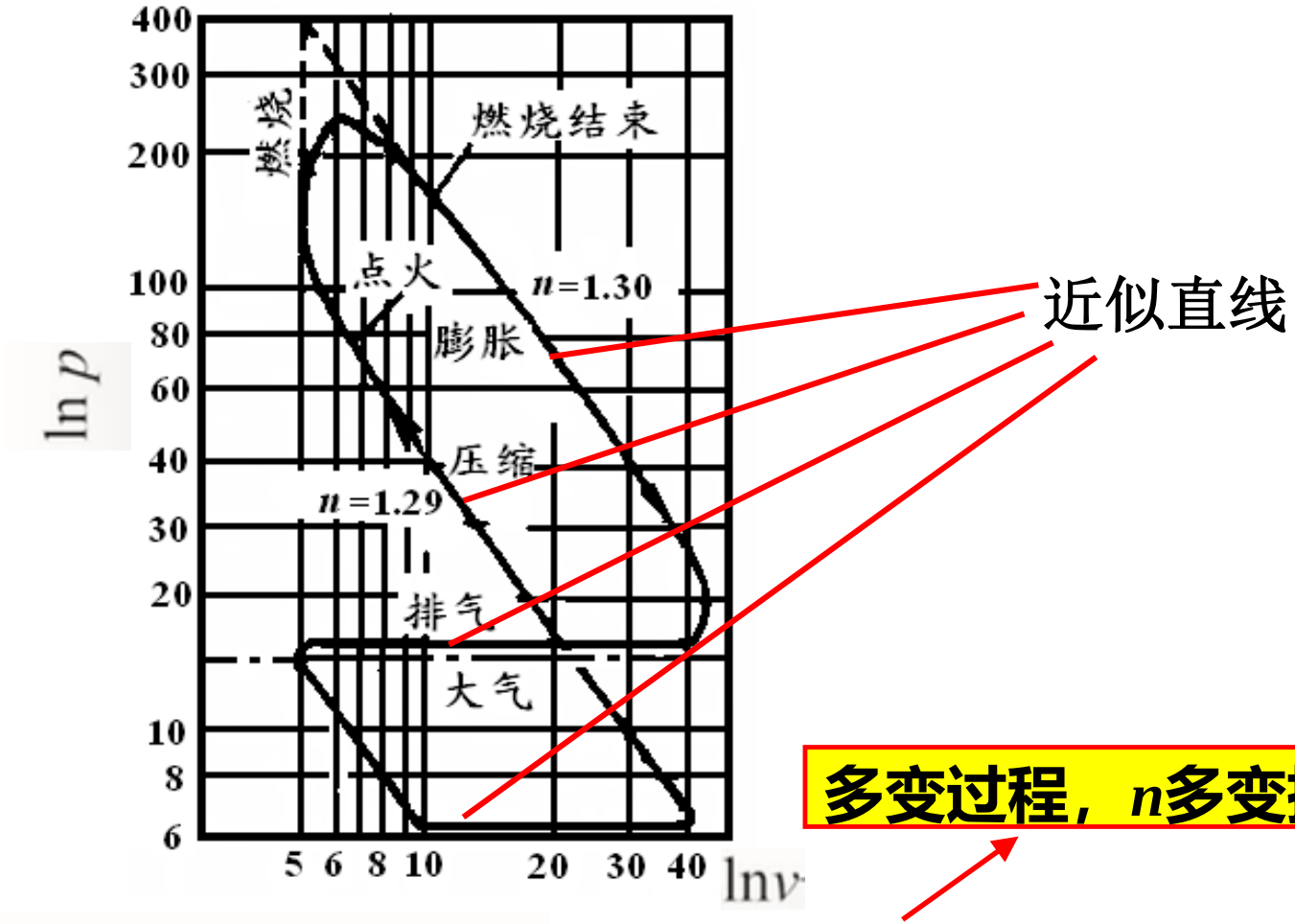
$$w_t \overset{\text{可逆}}{=\!=\!=} -\int v\mathrm{d}p=\begin{cases}-v\Delta p & \text{定容}\\ 0 & \text{定压}\end{cases}$$

$$w=q-\Delta u$$

$$w_t=q-\Delta h$$

二、基本热力过程 (fundamental thermodynamic process)

实测汽车发动机内工作过程燃气***p-V***关系



多变过程，*n*多变指数

在 p-V 图上有 $\ln p = -n\ln v + c$ $\Rightarrow$ $pv^n =$ **常数**

$n=0 \qquad p=\mathrm{const.}$ **定压过程**

(isobaric process; constant pressure process)

$n=1 \qquad pv=\mathrm{const.}$ **定温过程**

(isothermal process; constant temperature process)

$n=\kappa \qquad pv^{\kappa}=\mathrm{const.}$ **定熵（可逆绝热）过程**

(isentropic process; reversible adiabatic process)

$n=\pm\infty \qquad v=\mathrm{const.}$ **定容过程**

(isometric process; constant volume process)

pv^{n} **=常数 可逆多变过程 (polytropic process)**

4–2 理想气体的定压、定容和定温过程

一、定容过程—比体积不变的过程（v = 常数)

1、过程方程

$$n = \pm\infty \qquad v_1 = v_2$$

$$v_1 = \frac{R_g T_1}{p_1} \qquad v_2 = \frac{R_g T_2}{p_2} \qquad \boxed{\frac{p_1}{T_1} = \frac{p_2}{T_2}}$$

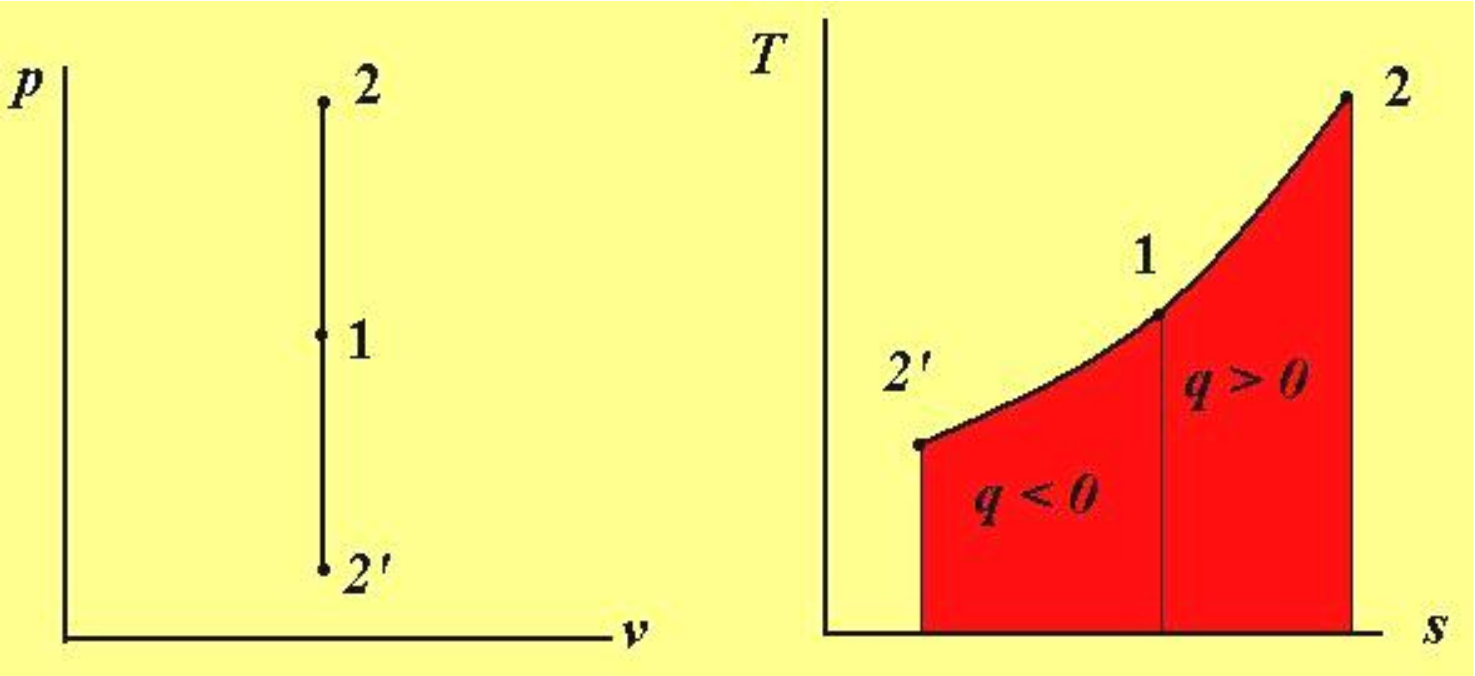
2、初、终态热学能、比焓、比熵的变化

$$\Delta u = u_2 - u_1 = c_v (T_2 - T_1)$$

$$\Delta h = h_2 - h_1 = c_p (T_2 - T_1)$$

$$\Delta s = c_v \ln \frac{T_2}{T_1} + R_g \ln \frac{v_2}{v_1} = c_v \ln \frac{T_2}{T_1}$$

3、在*p-v*、*T-s*图上表示



4、膨胀功、技术功和热量

$$w=\int_1^2 pdv=0 \qquad w_t=-\int_1^2 vdp=-v(p_2-p_1)$$

$$q_v=\Delta u=c_v(T_2-T_1)$$

与外界交换的热量未转变为机械能，全部用来改变热力学能。

二、定压过程—压力不变的过程（p = 常数)

1、过程方程

$$n = 0 \qquad p_1 = p_2$$

$$p_1 = \frac{R_g T_1}{v_1} \qquad p_2 = \frac{R_g T_2}{v_2} \qquad \boxed{\frac{v_1}{T_1} = \frac{v_2}{T_2}}$$

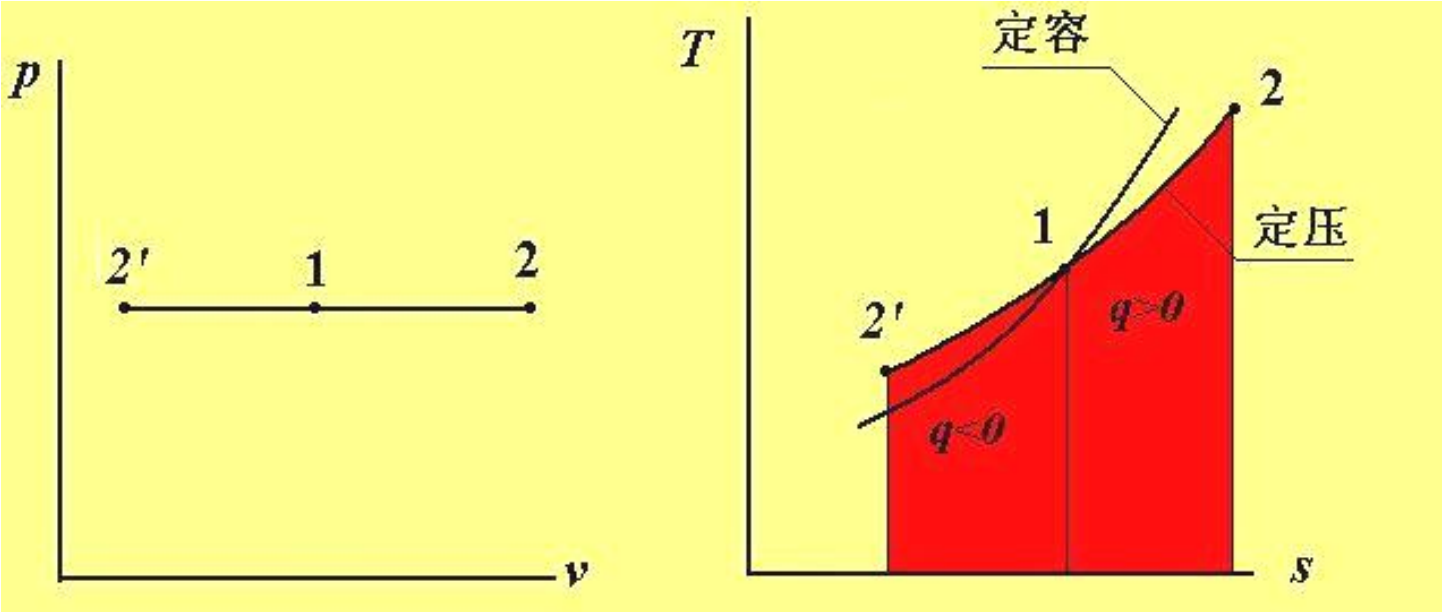
2、初、终态热学能、比焓、比熵的变化

$$\Delta u = u_2 - u_1 = c_v (T_2 - T_1)$$

$$\Delta h = h_2 - h_1 = c_p (T_2 - T_1)$$

$$\Delta s = c_p \ln \frac{T_2}{T_1} - R_g \ln \frac{p_2}{p_1} = c_p \ln \frac{T_2}{T_1}$$

3、在*p-v*、*T-s*图上表示



4、膨胀功、技术功和热量

$$w=\int_1^2 pdv=p(v_2-v_1) \qquad w_t=-\int_1^2 vdp=0$$

$$q_p=\Delta h=c_p(T_2-T_1)$$

热量转化来的机械能全部用来维持工质流动。

三、定温过程—温度不变的过程（T = 常数)

1、过程方程

$$n = 1 \qquad T_1 = T_2$$

$$T_1 = \frac{p_1 v_1}{R_g} \qquad T_2 = \frac{p_2 v_2}{R_g} \qquad \boxed{p_1 v_1 = p_2 v_2}$$

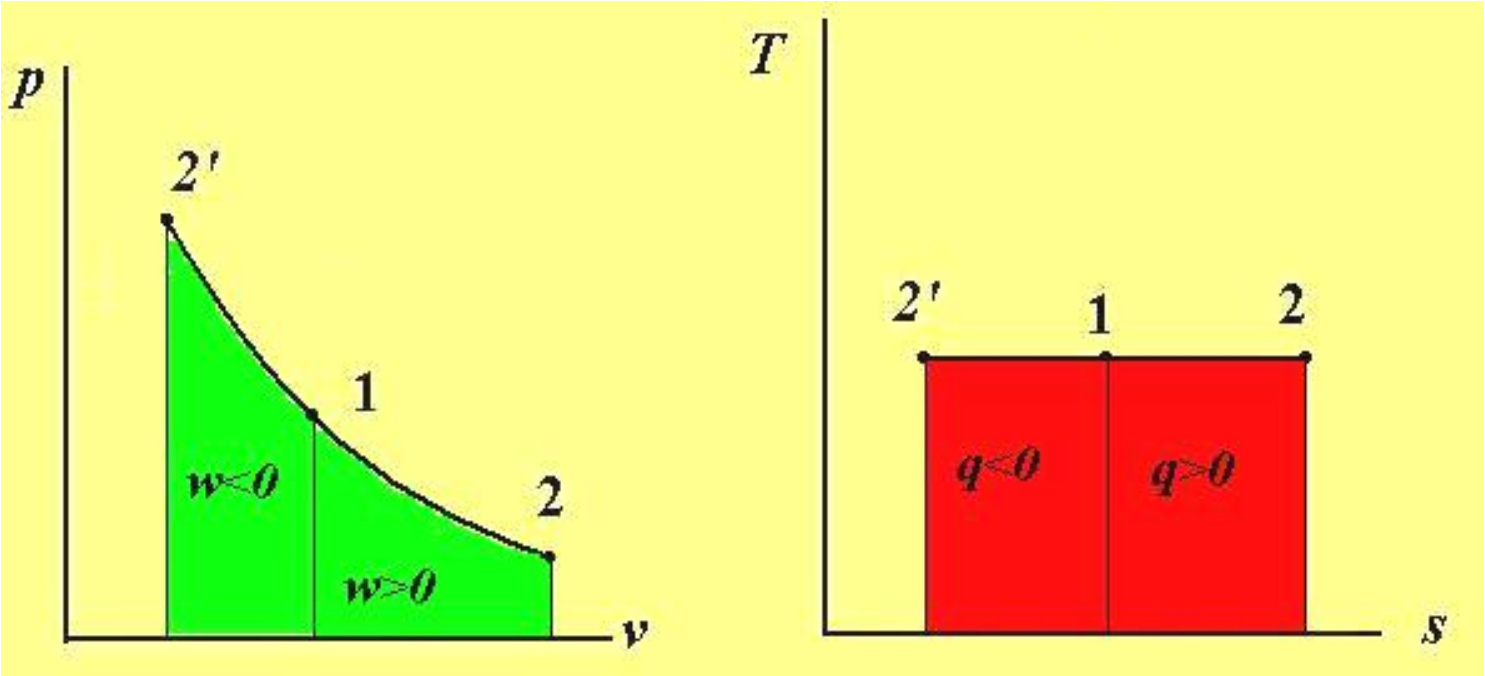
2、初、终态热学能、比焓、比熵的变化

$$\Delta u = u_2 - u_1 = 0$$

$$\Delta h = h_2 - h_1 = 0$$

$$\Delta s = -R_g \ln \frac{p_2}{p_1} = R_g \ln \frac{v_2}{v_1}$$

3、在*p*-*v*、*T*-*s*图上表示



4、膨胀功、技术功和热量

$$w=\int_1^2 pdv=R_gT\ln\frac{v_2}{v_1} \qquad w_t=-\int_1^2 vdp=-R_gT\ln\frac{p_2}{p_1}$$

$$w=w_t=R_gT\ln\frac{v_2}{v_1}=-R_gT\ln\frac{p_2}{p_1}$$

$$\boxed{q_T=w=w_t=T(s_2-s_1)}$$

4–3 理想气体定熵（可逆绝热）过程

$$\delta q = 0, \quad \Rightarrow \mathrm{d}s = \left(\frac{\delta q}{T}\right)_{\mathrm{re}} = 0, \quad s = \mathrm{const}$$

定熵过程 = 绝热过程 + 可逆过程

一、过程方程

$$T\mathrm{d}s = \delta q = \mathrm{d}h - v\mathrm{d}p = 0 \Rightarrow \qquad v\mathrm{d}p = \mathrm{d}h = c_p \mathrm{d}T \qquad (A)$$

$$T\mathrm{d}s = \delta q = \mathrm{d}u + p\mathrm{d}v = 0 \Rightarrow \qquad -p\mathrm{d}v = \mathrm{d}u = c_V \mathrm{d}T \qquad (B)$$

$$(A) \div (B) \qquad \Rightarrow \frac{c_p}{c_v} = -\frac{vdp}{pdv}$$

令 $\gamma = \dfrac{c_p}{c_v}$ $\qquad \dfrac{dp}{p} + \gamma \dfrac{dv}{v} = 0 \qquad \ln p + \gamma \ln v = \mathrm{C} \qquad pv^{\gamma} = 定值$

定熵过程方程 $\qquad pv^{\kappa} = 定值$

理想气体定熵过程的定熵指数等于比热比。

二、初、终态参数间的关系

$$\boxed{p_1 v_1^{\kappa} = p_2 v_2^{\kappa}}$$

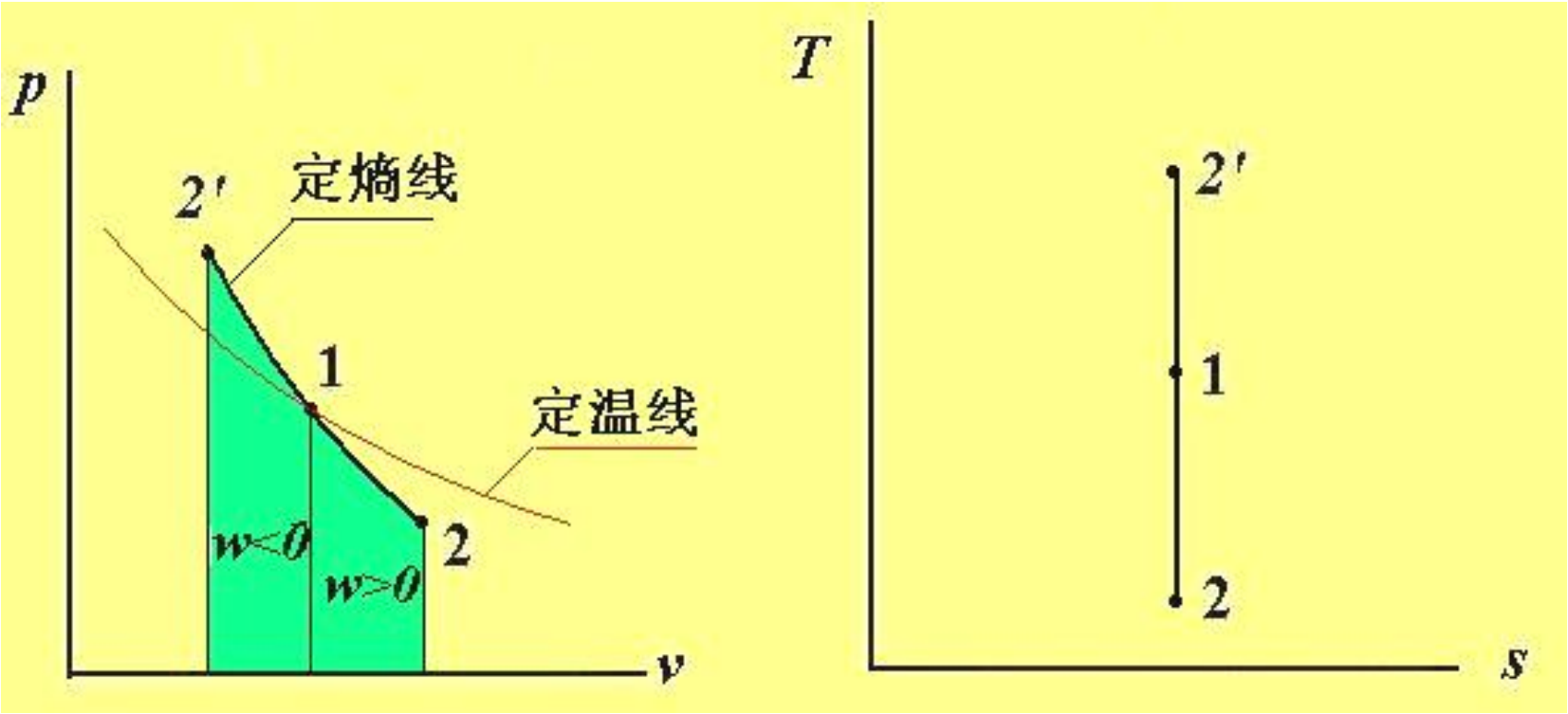
$$\boxed{T_1 v_1^{\kappa-1} = T_2 v_2^{\kappa-1}}$$

$$\boxed{T_1 p_1^{-\frac{\kappa-1}{\kappa}} = T_2 p_2^{-\frac{\kappa-1}{\kappa}}}$$

上述三式适用于：

理想气体，定比热，可逆绝热过程。

三、过程的 $p-v$ 图及 $T-s$ 图



定熵过程 $\left(\frac{\partial p}{\partial v}\right)_s = -\kappa \frac{p}{v}$　　定温过程 $\left(\frac{\partial p}{\partial v}\right)_T = -\frac{p}{v}$

$$\kappa > 1$$

四、过程中能量的传递和转换

$$q = 0$$

$$w = \underset{0}{\cancel{q}} - \Delta u = -\Delta u = u_1 - u_2 = c_V\left(T_1 - T_2\right) = \frac{R_g}{\kappa - 1}\left(T_1 - T_2\right)$$

$$= \frac{R_g T_1}{\kappa - 1}\left[1 - \left(\frac{T_2}{T_1}\right)\right] = \frac{R_g T_1}{\kappa - 1}\left[1 - \left(\frac{p_2}{p_1}\right)^{\frac{\kappa - 1}{\kappa}}\right]$$

比较

$$w_t = \underset{0}{\cancel{q}} - \Delta h = -\Delta h = h_1 - h_2 = c_p\left(T_1 - T_2\right) = \frac{\kappa}{\kappa - 1} R_g\left(T_1 - T_2\right) = \kappa w$$

$$\boxed{w_t = \kappa w}$$

4–4 理想气体热力过程综合分析

多变过程方程式为 pv^n=常数，n取不同值，可得到不同的过程。

当$n = 0$时，p = 常数，则为定压过程

当$n = 1$时，pv = 常数，则为定温过程

当$n = \kappa$时，pv^κ = 常数，则为定熵过程

当$n = \pm\infty$时，v = 常数，则为定容过程

因此，把多变过程表示在p-v图上时，要先画出四个基本热力过程，然后再根据 n 值的大小画出相应的多变过程。

$$\boxed{\frac{dp}{dv} = -n\frac{p}{v}}$$

$$\boxed{\frac{dT}{ds} = \frac{T}{c_n} = \frac{T}{\frac{n-\kappa}{n-1}c_V}}$$

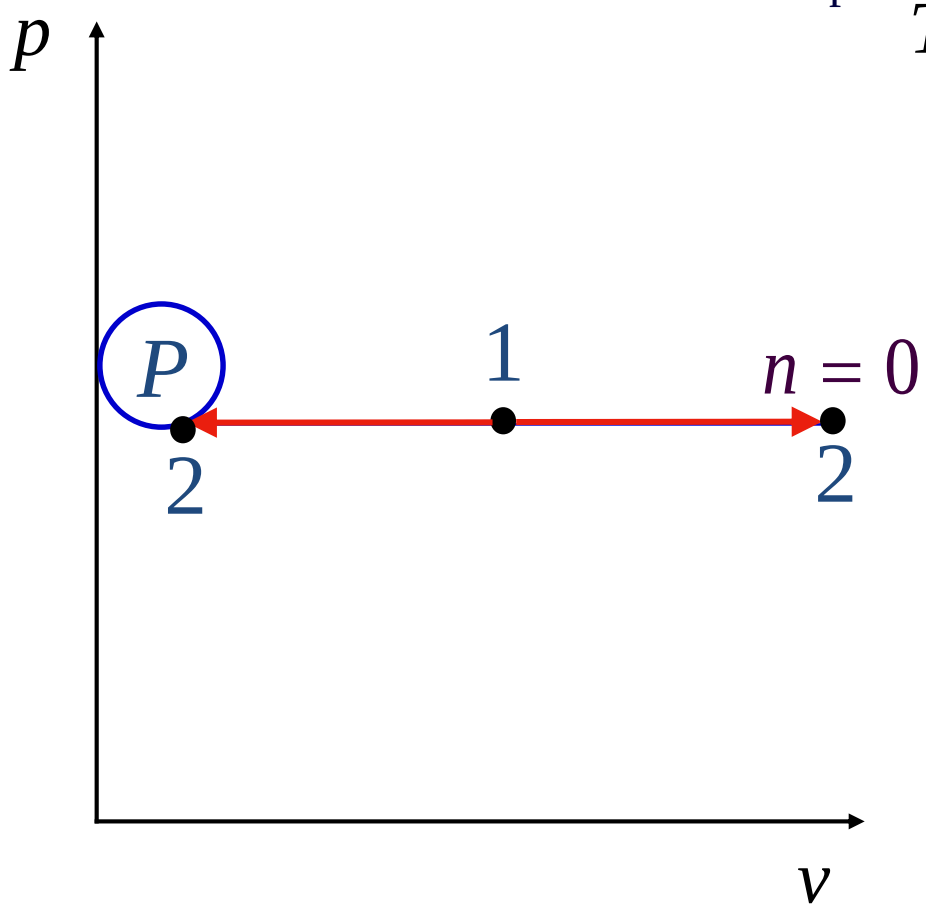
理想气体 P 过程的 p-v, T-s 图

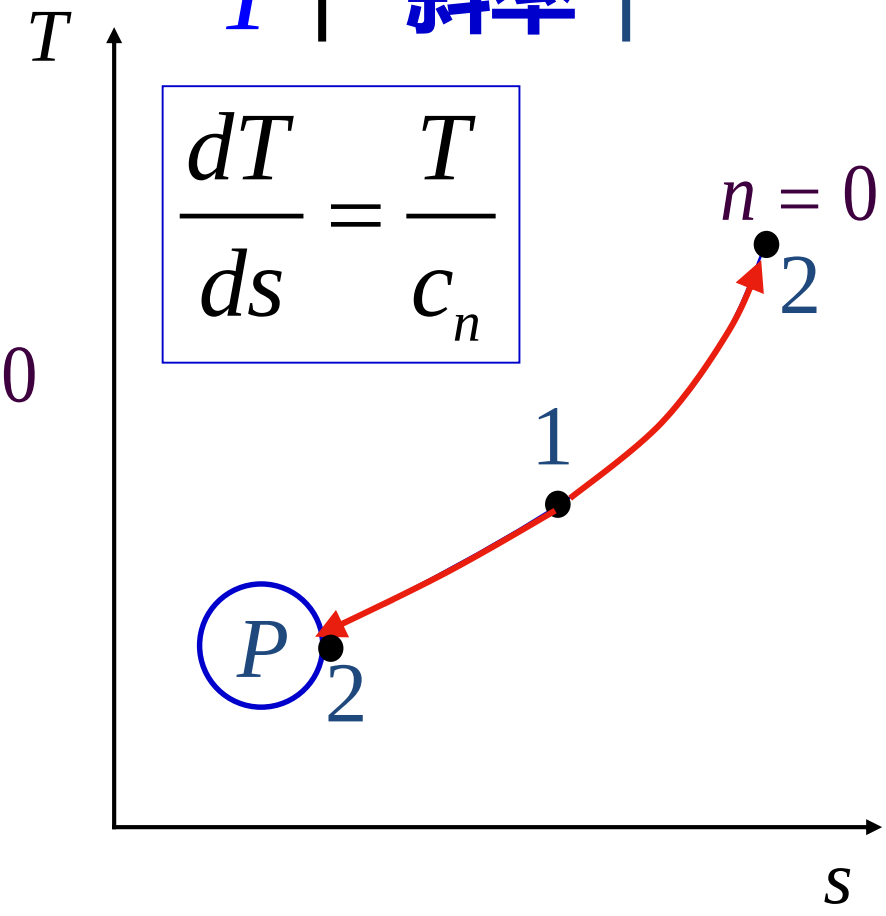
$$\left(\frac{dT}{ds}\right)_p = \frac{T}{c_p}$$

上凸?下凹?

T ↑ 斜率 ↑

$$\boxed{\frac{dT}{ds} = \frac{T}{c_n}}$$





理想气体 P 过程：

$$w_p = \int_1^2 p dv = p(v_2 - v_1)$$

$$w_{t,p} = -\int_1^2 v dp = 0$$

$$q_p = \int_1^2 T ds = \int_1^2 c_p dT$$

$$q_p = h_2 - h_1 + w_{t,p} = h_2 - h_1$$

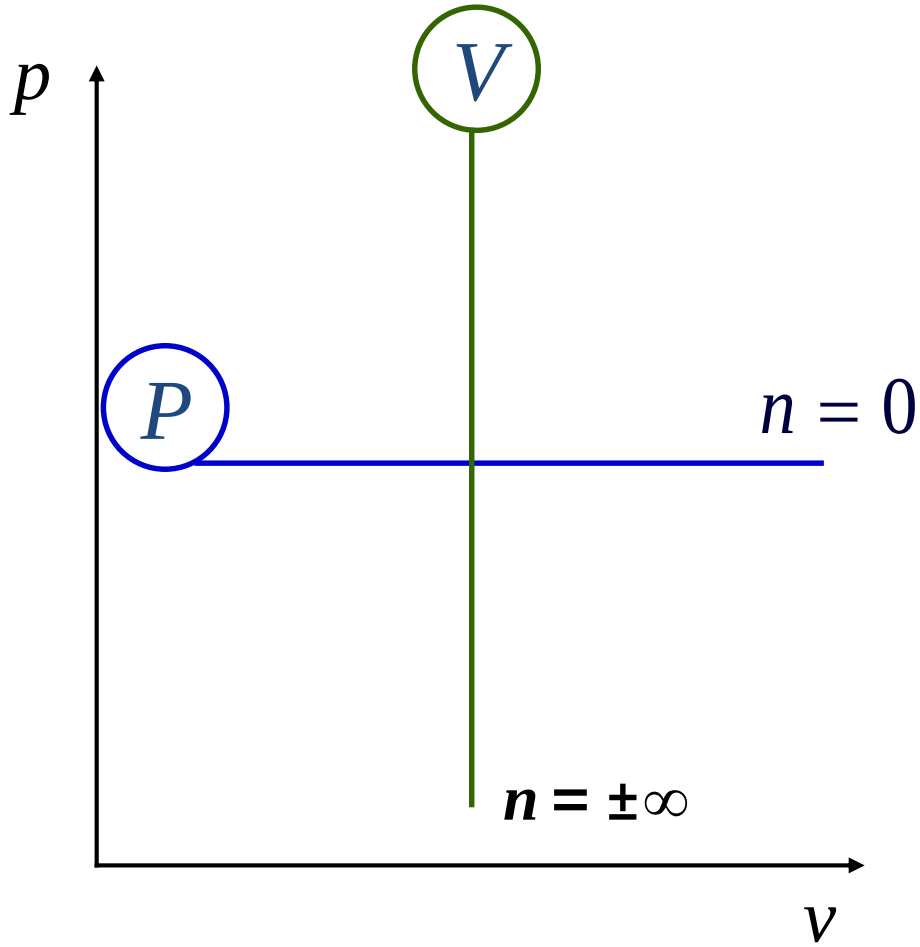
理想气体 V 过程的 p-v, T-s 图

$$\left(\frac{dT}{ds}\right)_v = \frac{T}{c_v}$$

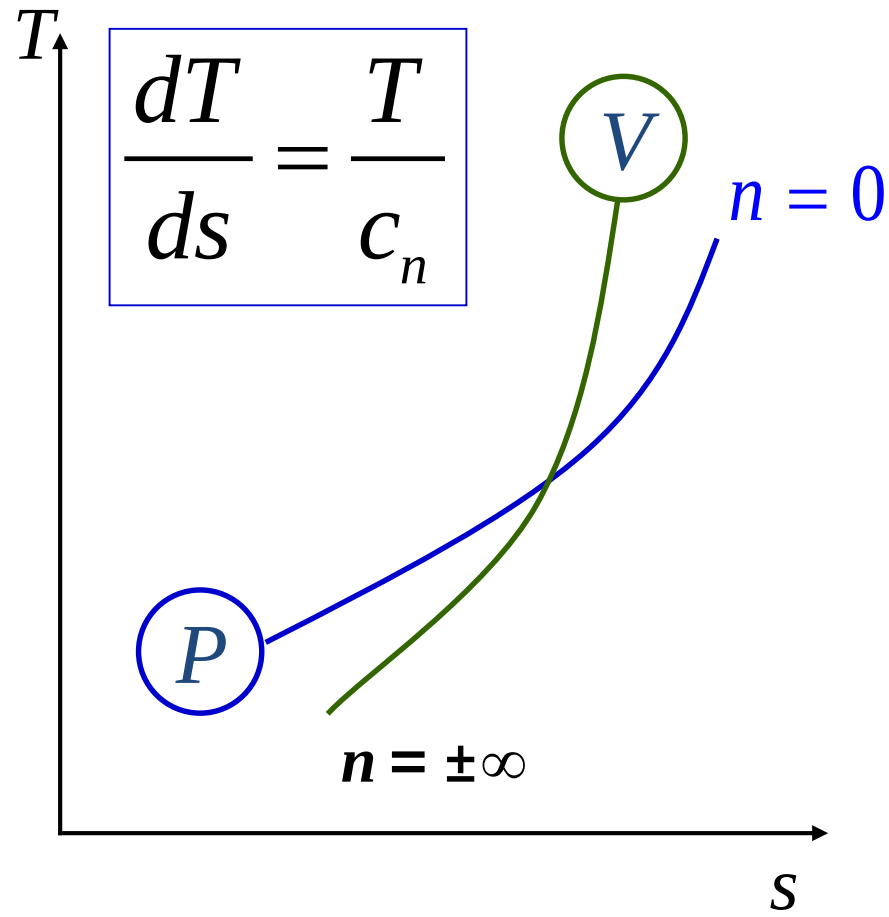
上凸? 下凹?

$c_p > c_v$

$T\uparrow$ 斜率 $\uparrow$



$$\boxed{\frac{dT}{ds} = \frac{T}{c_n}}$$



理想气体 V 过程：

$$w_v = \int_1^2 p dv = 0$$

$$w_{t,v} = -\int_1^2 v dp = v(p_1 - p_2)$$

$$q_v = \int_1^2 T ds = \int_1^2 c_v dT$$

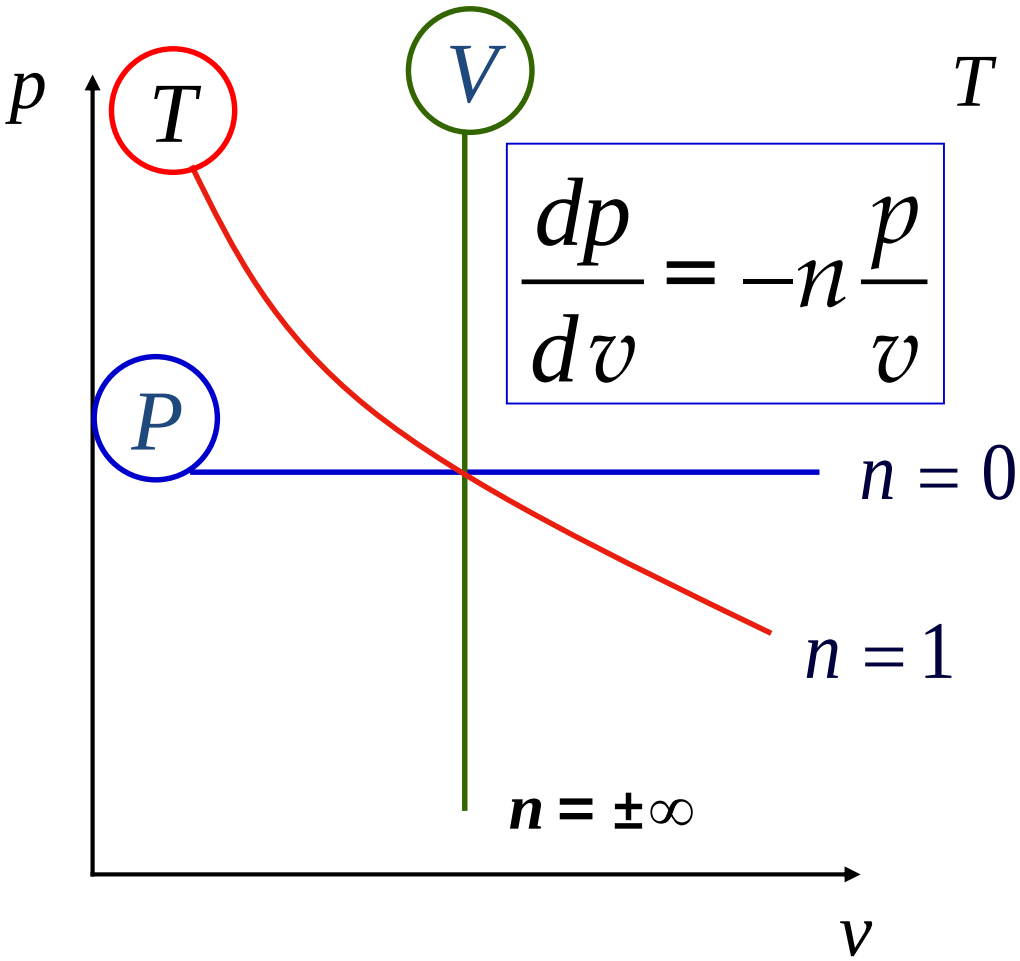
$$q_v = u_2 - u_1 + w_v = u_2 - u_1$$

理想气体 T 过程的 p-v, T-s 图

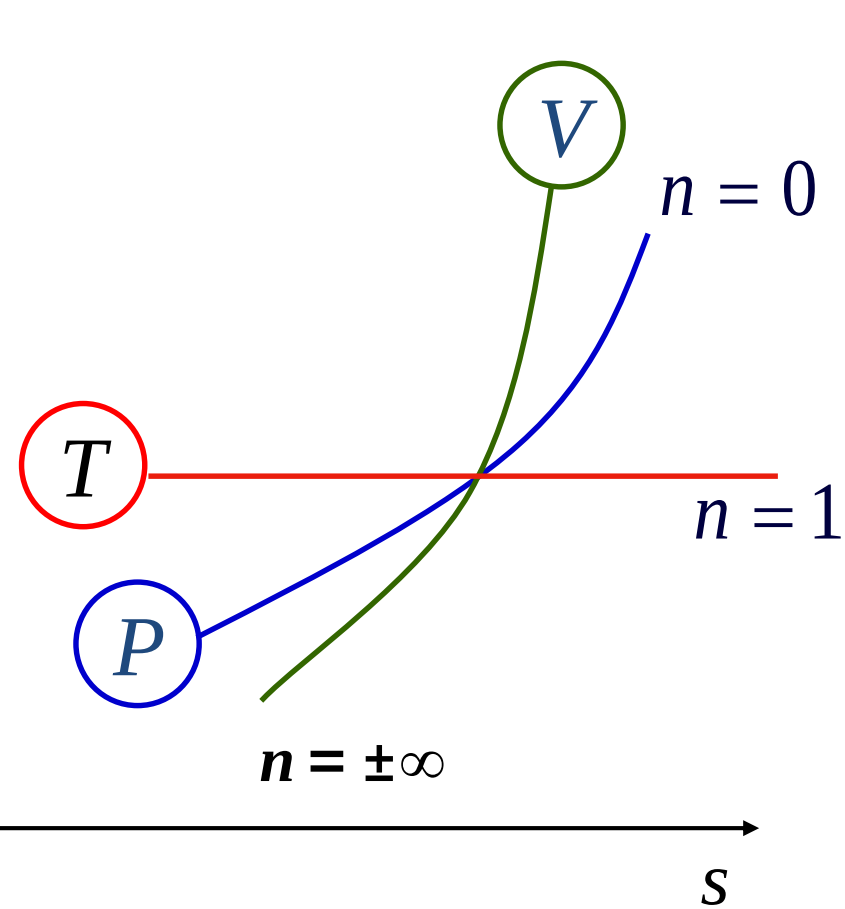
T $\left(\frac{dp}{dv}\right)_T = -\frac{p}{v}$

上凸? 下凹?

p↑ 斜率↑



$$\frac{dp}{dv} = -n\frac{p}{v}$$



理想气体 T 过程：

$$w_T = \int_1^2 p dv = \int_1^2 \frac{R_g T}{v} dv = R_g T \ln \frac{v_2}{v_1}$$

$$w_{t,T} = -\int_1^2 v dp = -\int_1^2 \frac{R_g T}{p} dp = R_g T \ln \frac{p_1}{p_2}$$

$$\frac{v_2}{v_1} = \frac{p_1}{p_2} \quad \Rightarrow \quad w_{t,T} = w_t$$

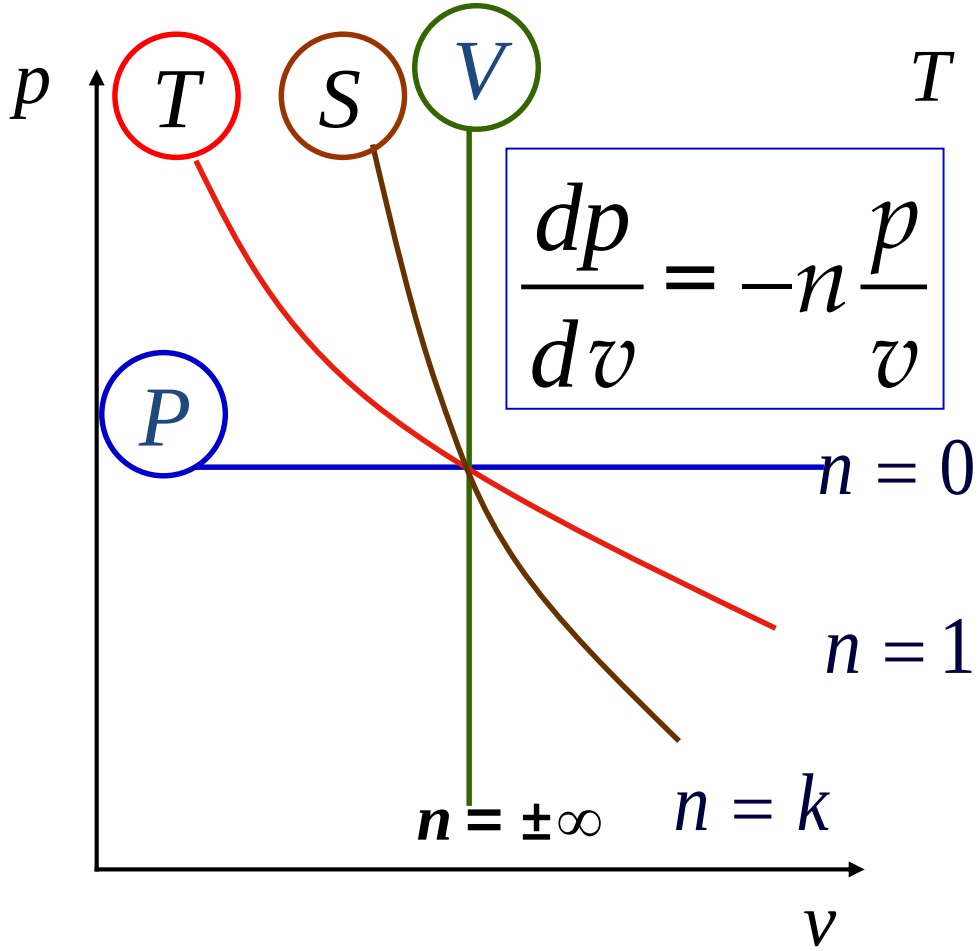
$$q_T = \int_1^2 T ds = T(s_2 - s_1)$$

$$q_T = u_2 - \cancelto{0}{u_1} + w_T = h_2 - \cancelto{0}{h_1} + w_{t,T}$$

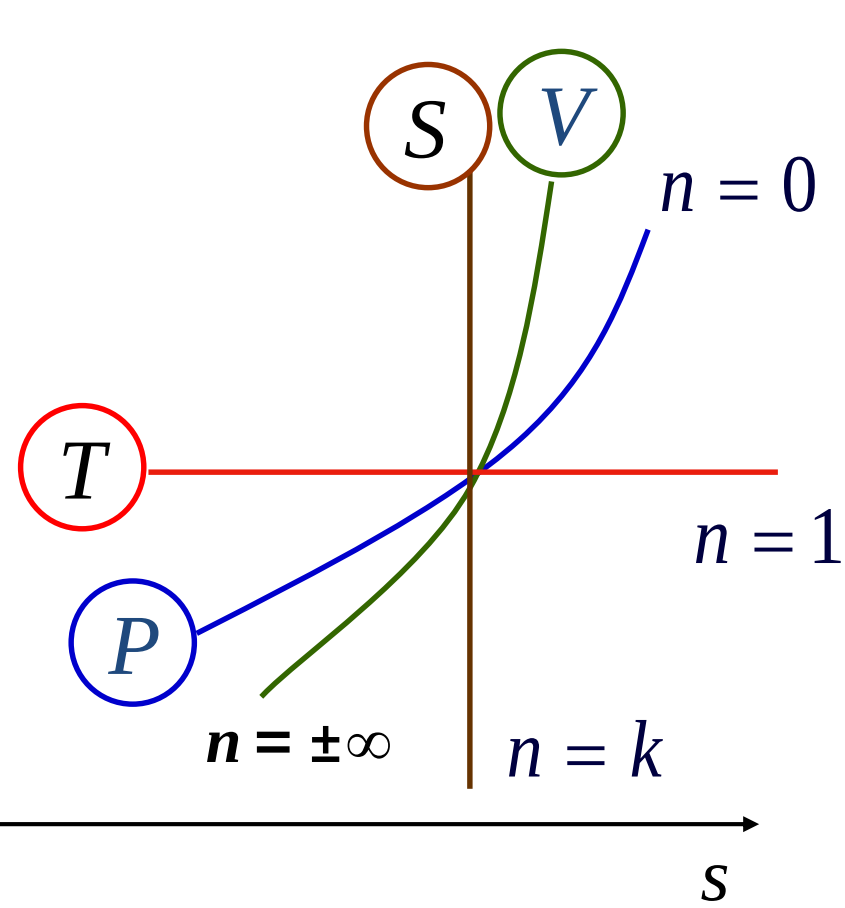
理想气体 s 过程的 p-v, T-s 图

$$\left(\frac{dp}{dv}\right)_s = -\frac{kp}{v}$$

$$\left(\frac{dp}{dv}\right)_T = -\frac{p}{v}$$



$$\frac{dp}{dv} = -n\frac{p}{v}$$



理想气体 s 过程：

$$w_{12} = q - \Delta u_{12} = u_1 - u_2$$

$$w_{t12} = q - \Delta h_{12} = h_1 - h_2$$

$$w_{12} = c_v(T_1 - T_2) = \frac{R_g}{k-1}(T_1 - T_2) = \frac{R_g T_1}{k-1}(1 - \frac{T_2}{T_1}) = \frac{1}{k-1}(P_1V_1 - P_2V_2)$$

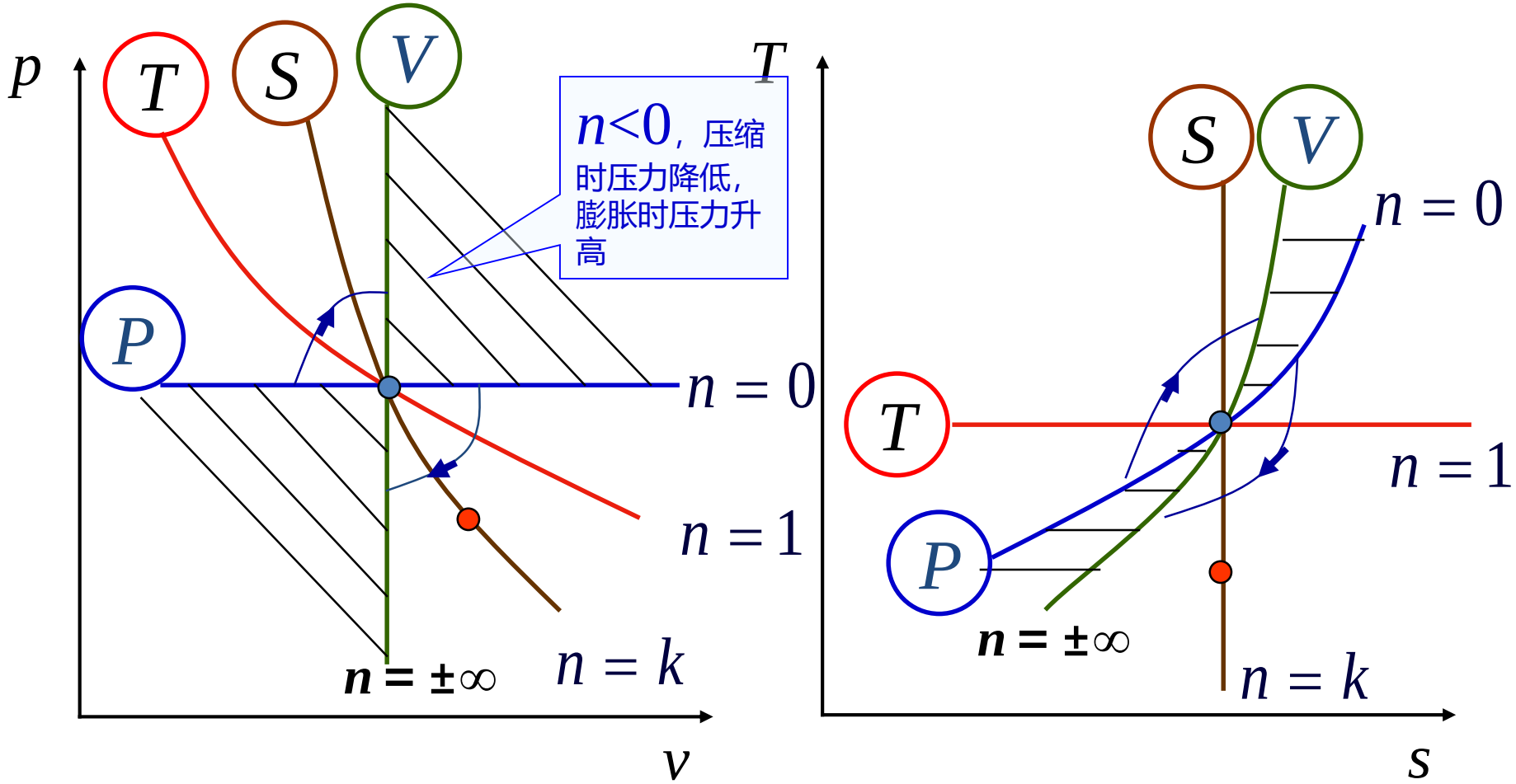
$$w_{t12} = c_p(T_1 - T_2) = \frac{kR_g}{k-1}(T_1 - T_2) = \frac{kR_g T_1}{k-1}(1 - \frac{T_2}{T_1}) = \frac{k}{k-1}(P_1V_1 - P_2V_2)$$

$$w_{t12} = kw_{12}$$

$$q = 0$$

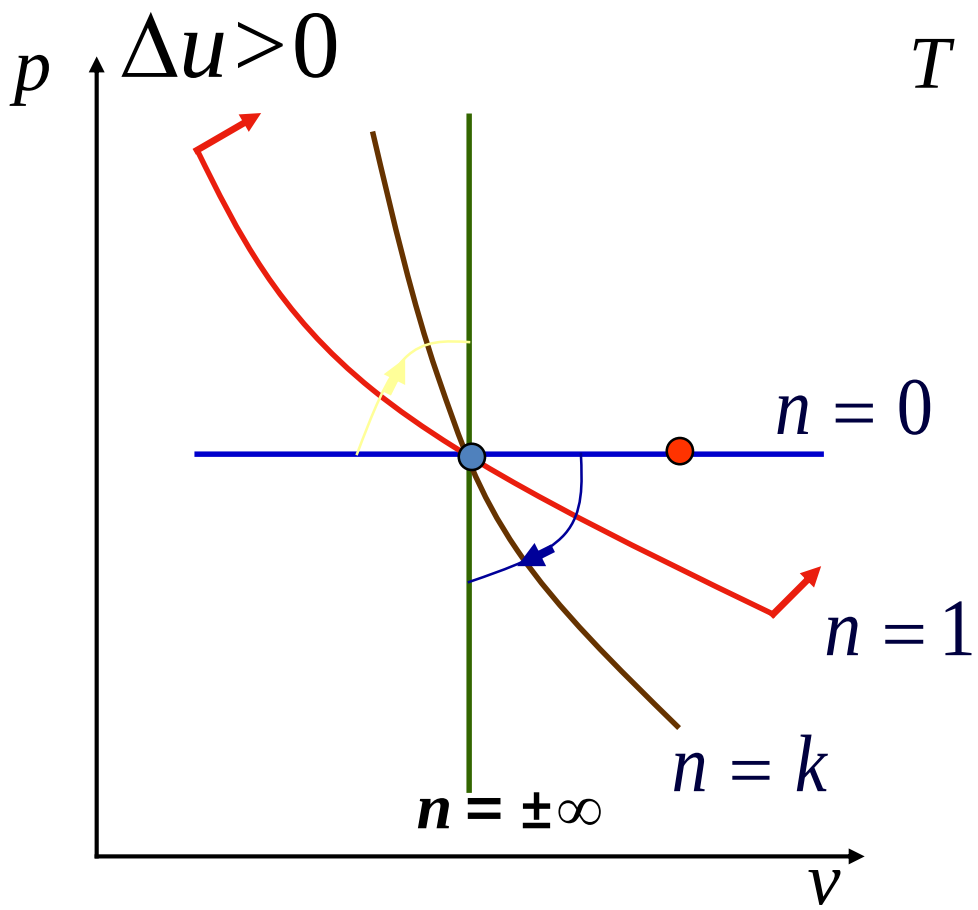
理想气体基本过程的p-v,T-s图

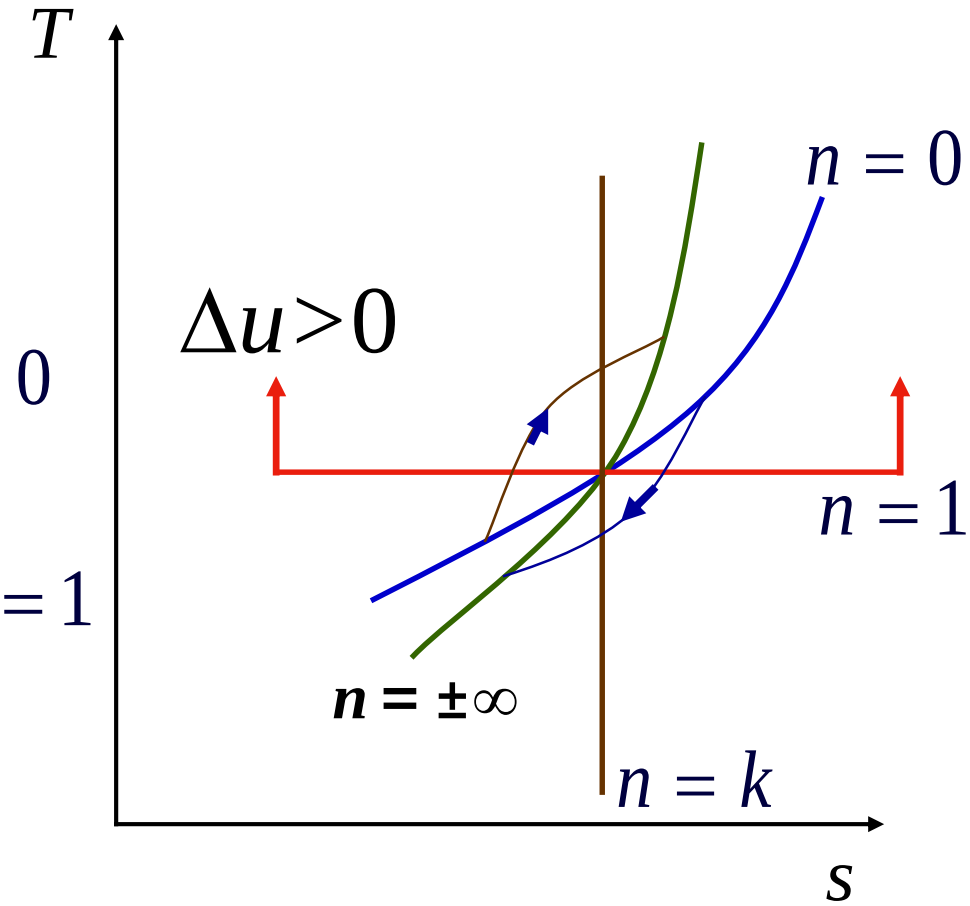
$$\frac{T_2}{T_1}=\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}$$



u 在p-v,T-s图上的变化趋势

$u = T$ $\uparrow \Delta u = \int c_v dT \uparrow$ $pv = R_g T$

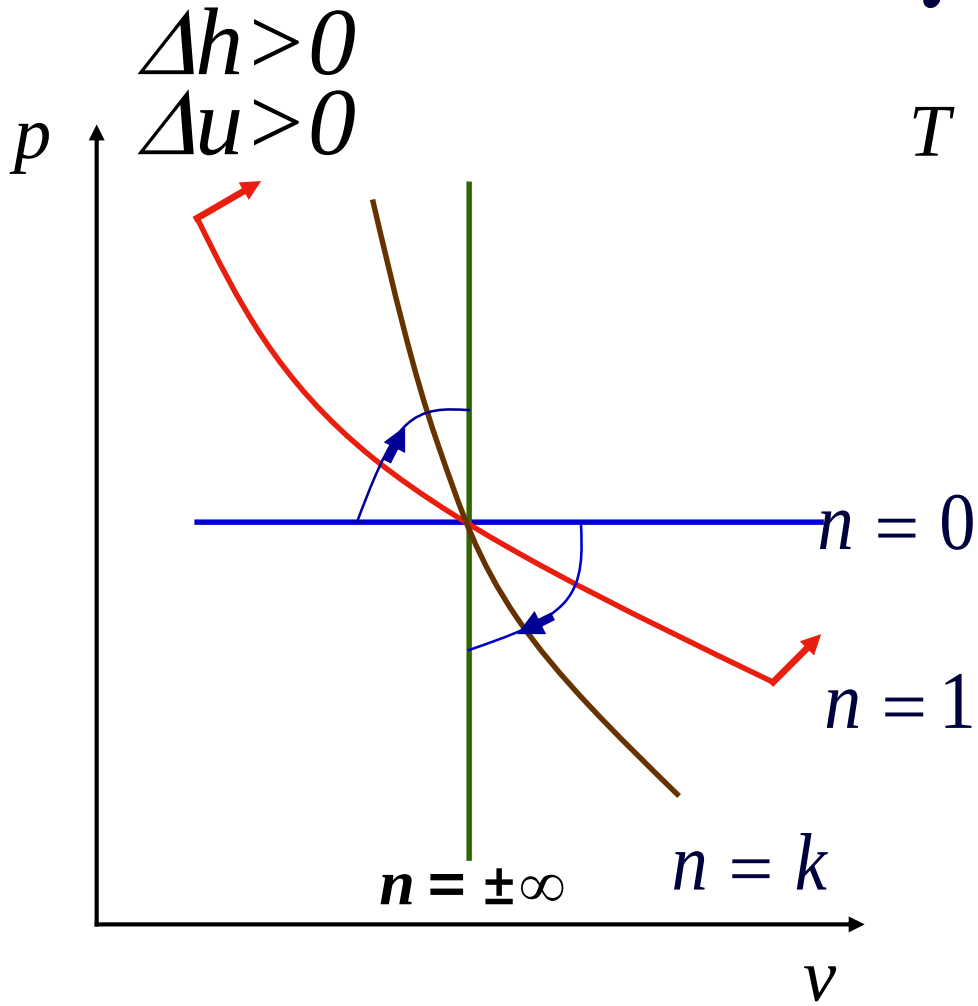


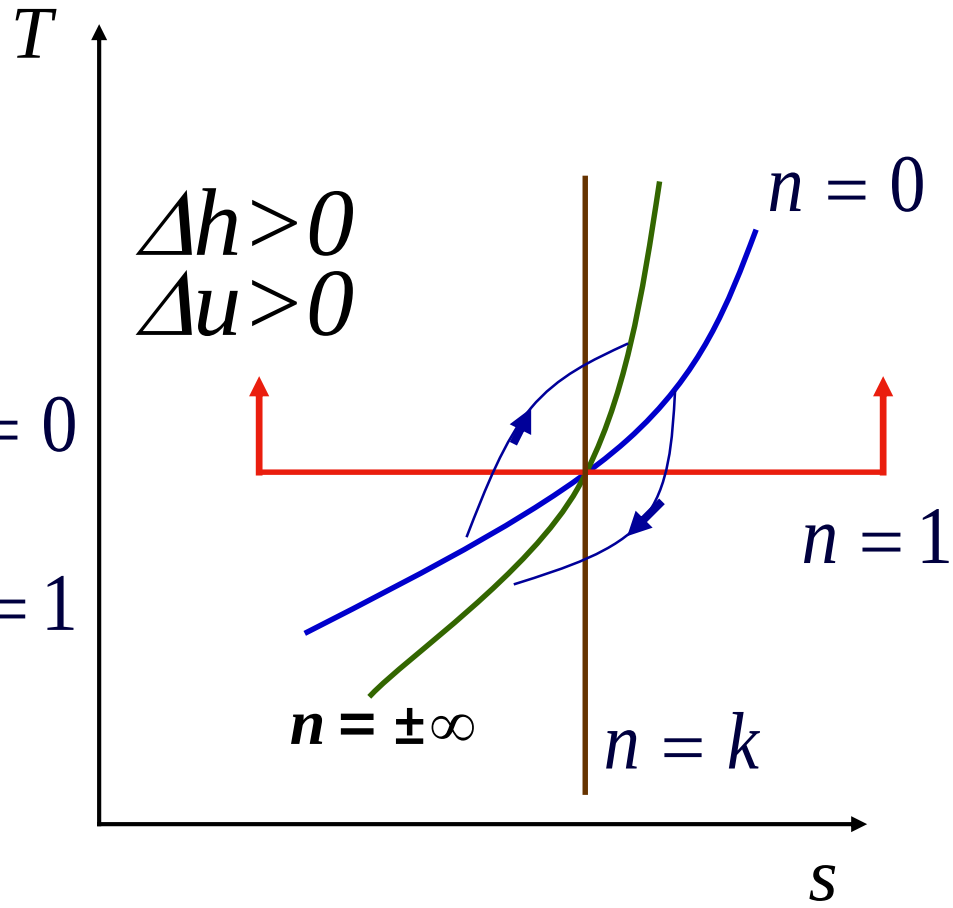


h 在p-v,T-s图上的变化趋势

$h = T$

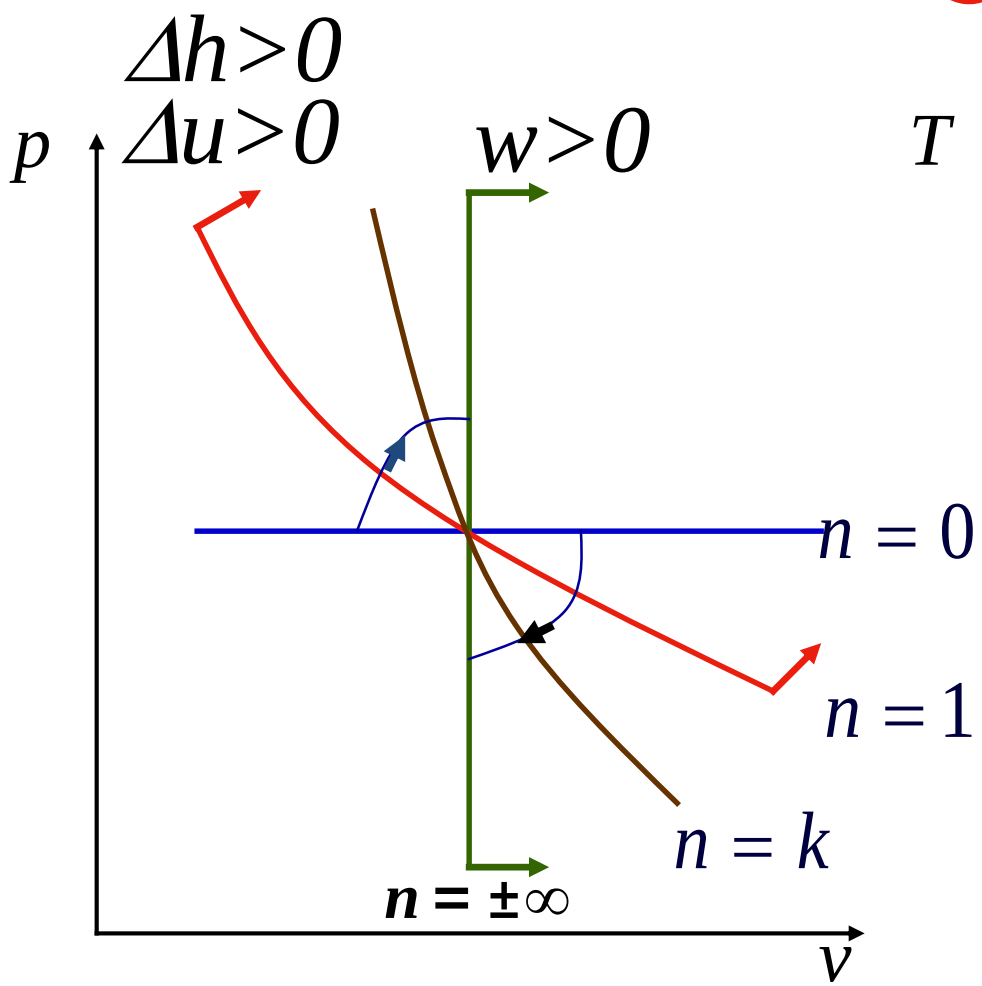
$\uparrow \Delta h = \int c_p dT \uparrow$

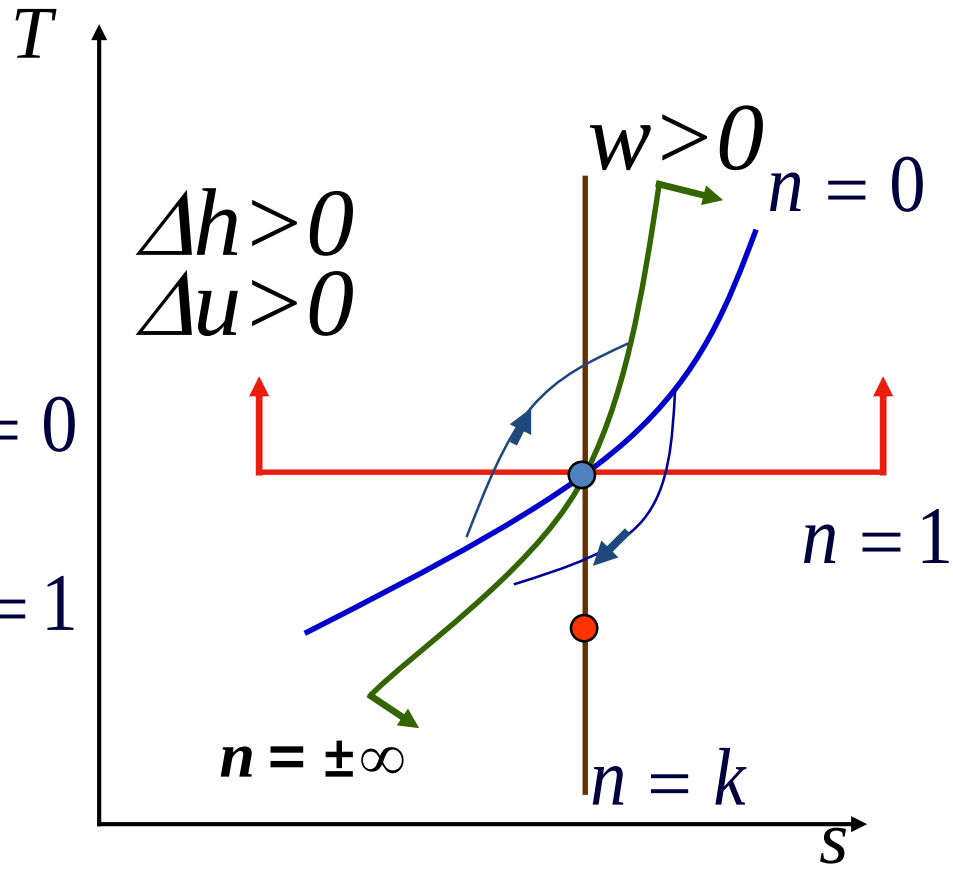




w 在 p-v, T-s 图上的变化趋势

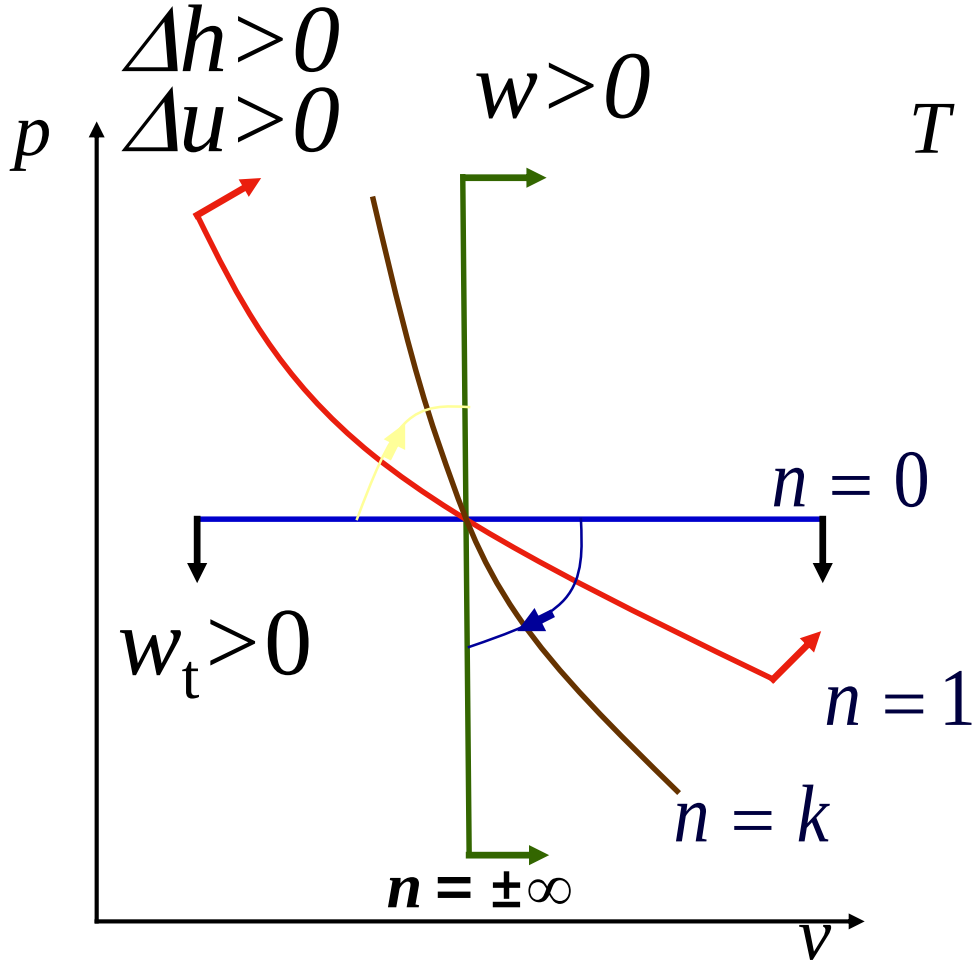
$$\uparrow w = \int p\,dv \uparrow \qquad \frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{k-1}$$

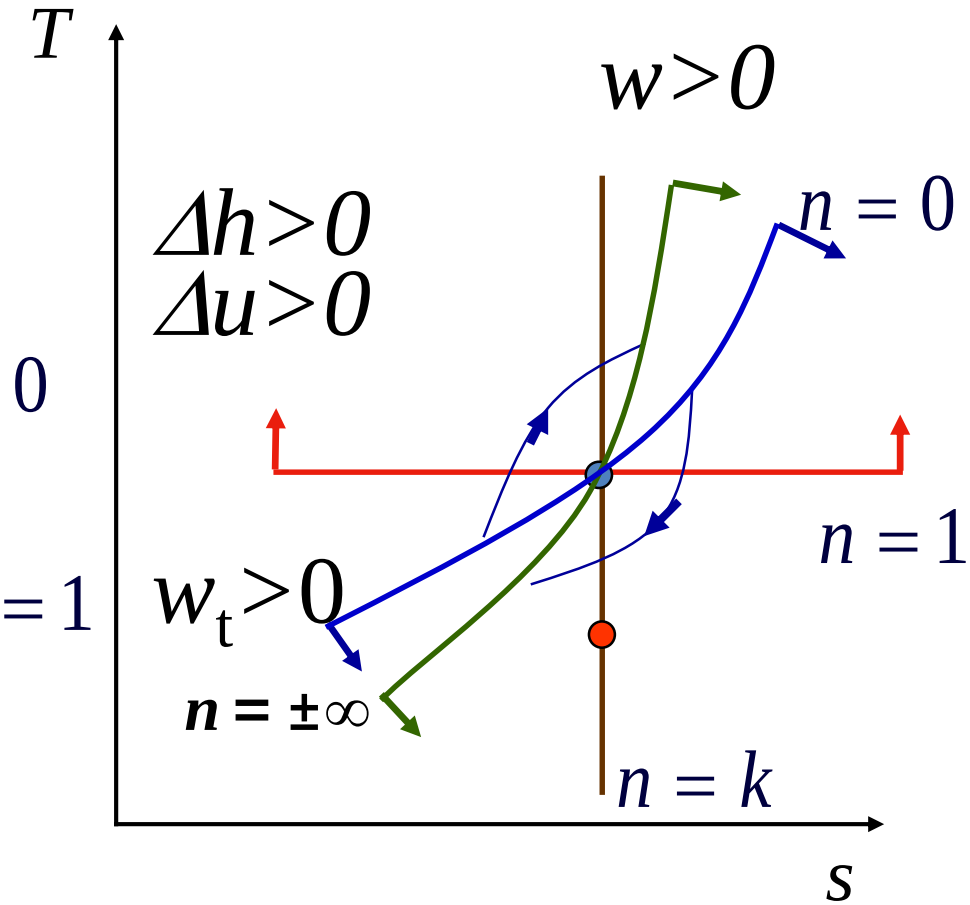




w_t在p-v,T-s图上的变化趋势

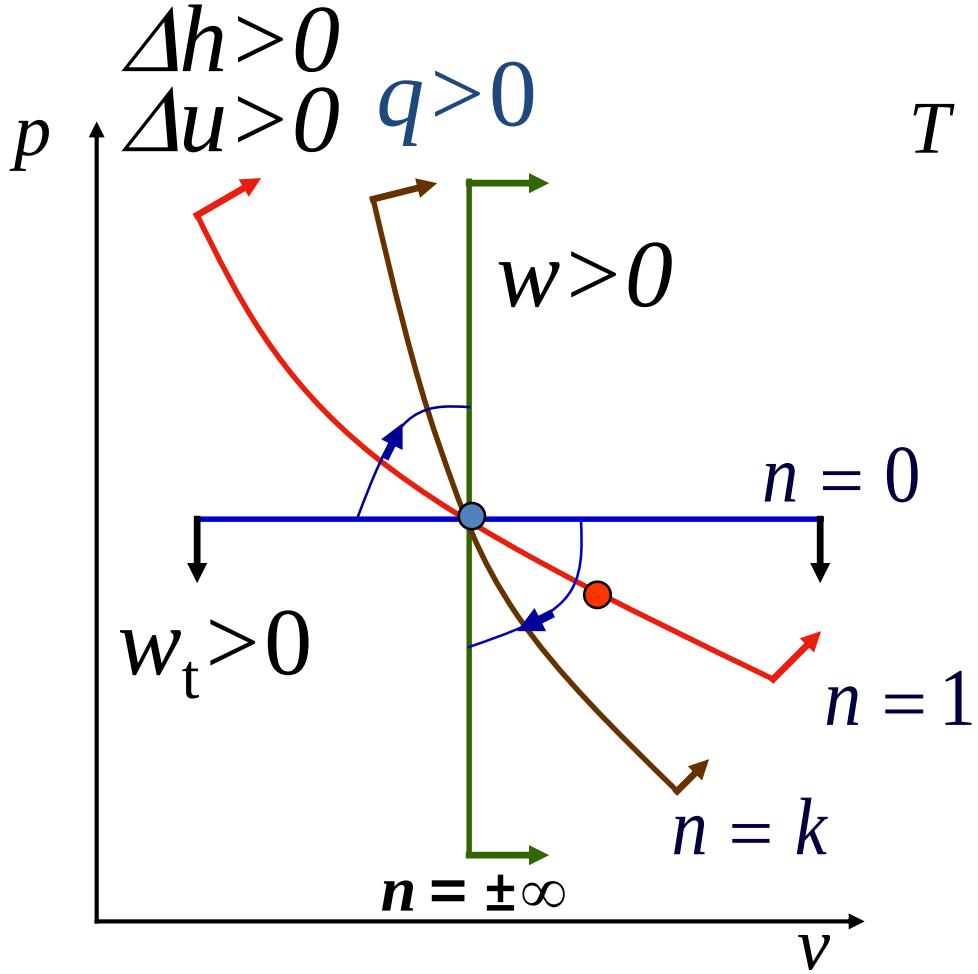
$$\uparrow w_t = -\int v\,dp \downarrow \qquad \frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}}$$

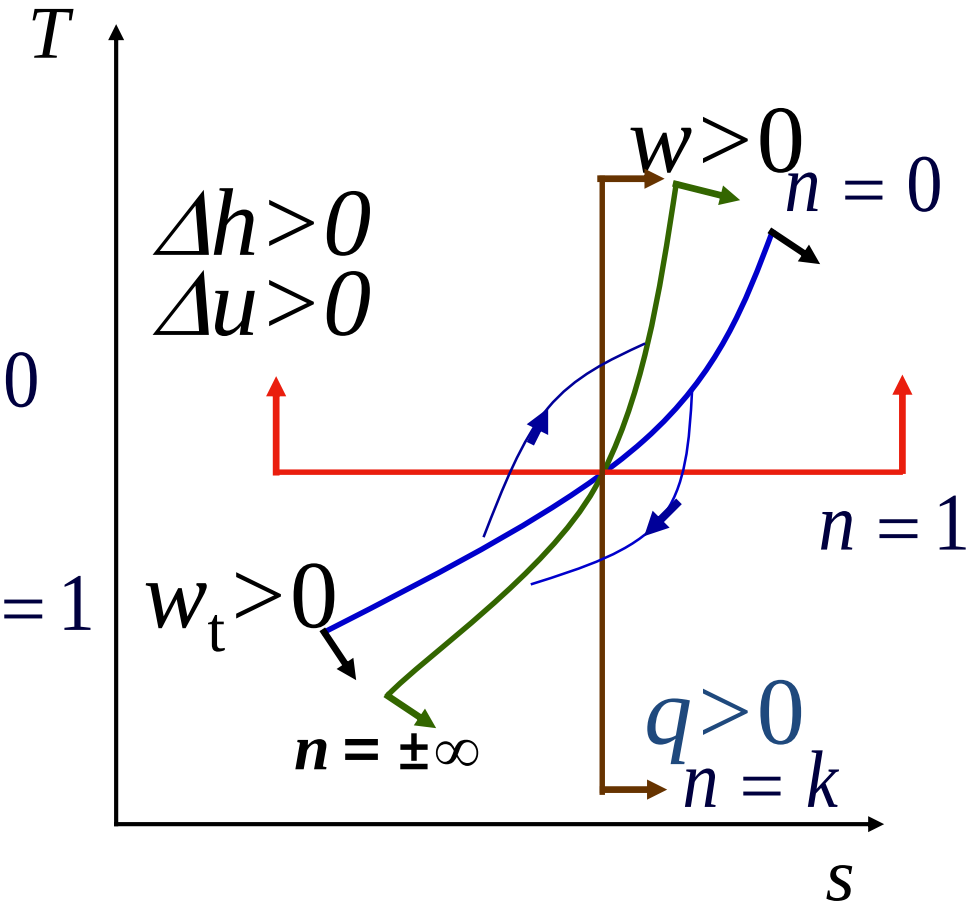




q在p-v,T-s 图上的变化趋势

$$\uparrow q = \int T ds \uparrow \qquad T \qquad q = w$$

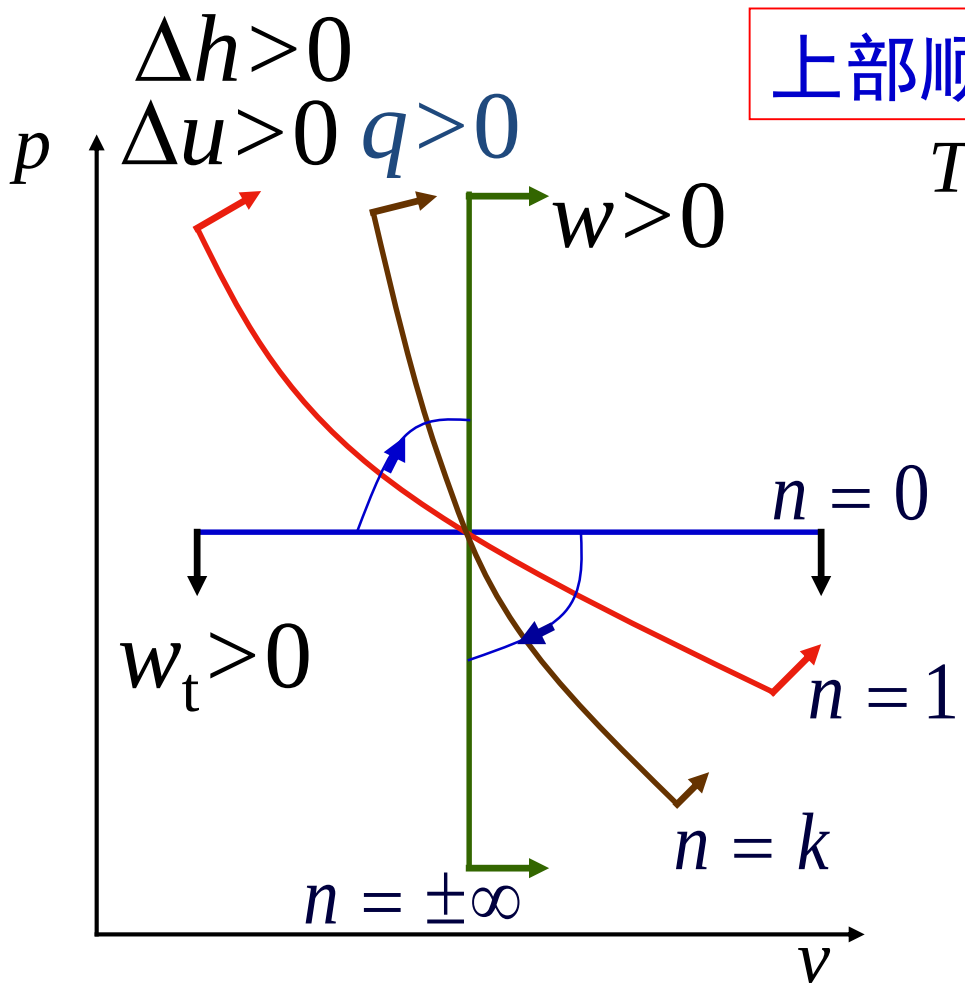


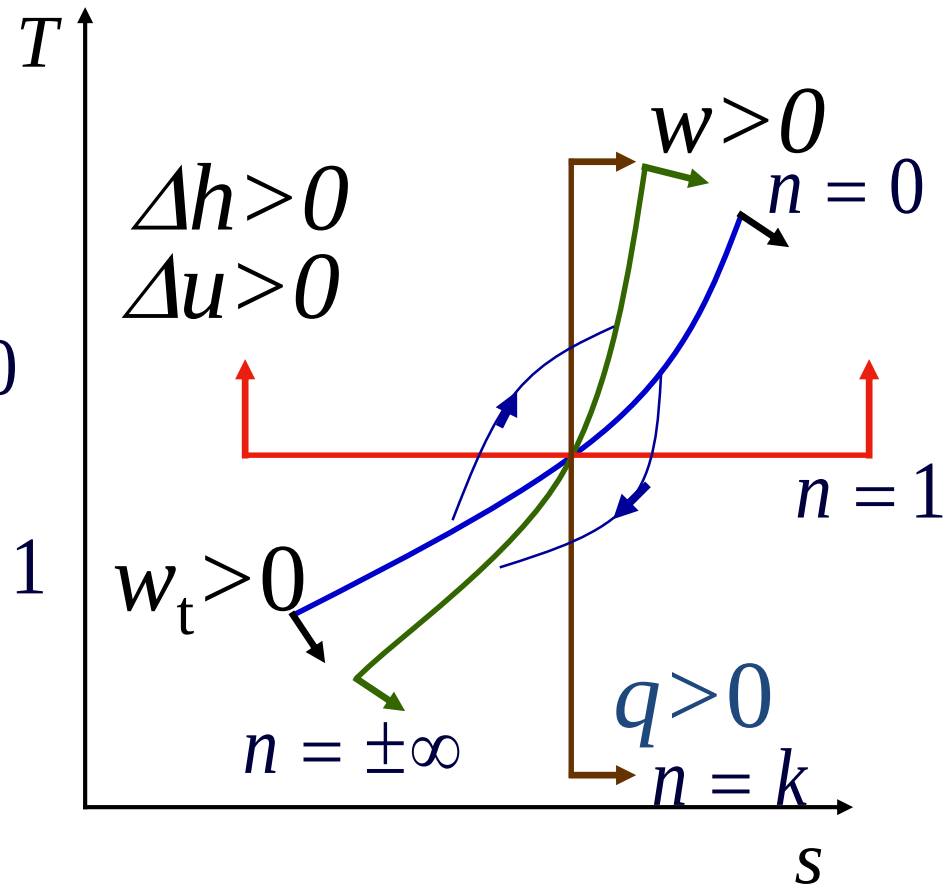


u,h,w,w_t,q在p-v,T-s图上的变化趋势

$u,h\uparrow(T\uparrow)$ $w\uparrow(v\uparrow)$ $w_t\uparrow(p\downarrow)$ $q\uparrow(s\uparrow)$

上部顺时针方向变化





分界线

等温线是**热力学能**和**焓**的分界线

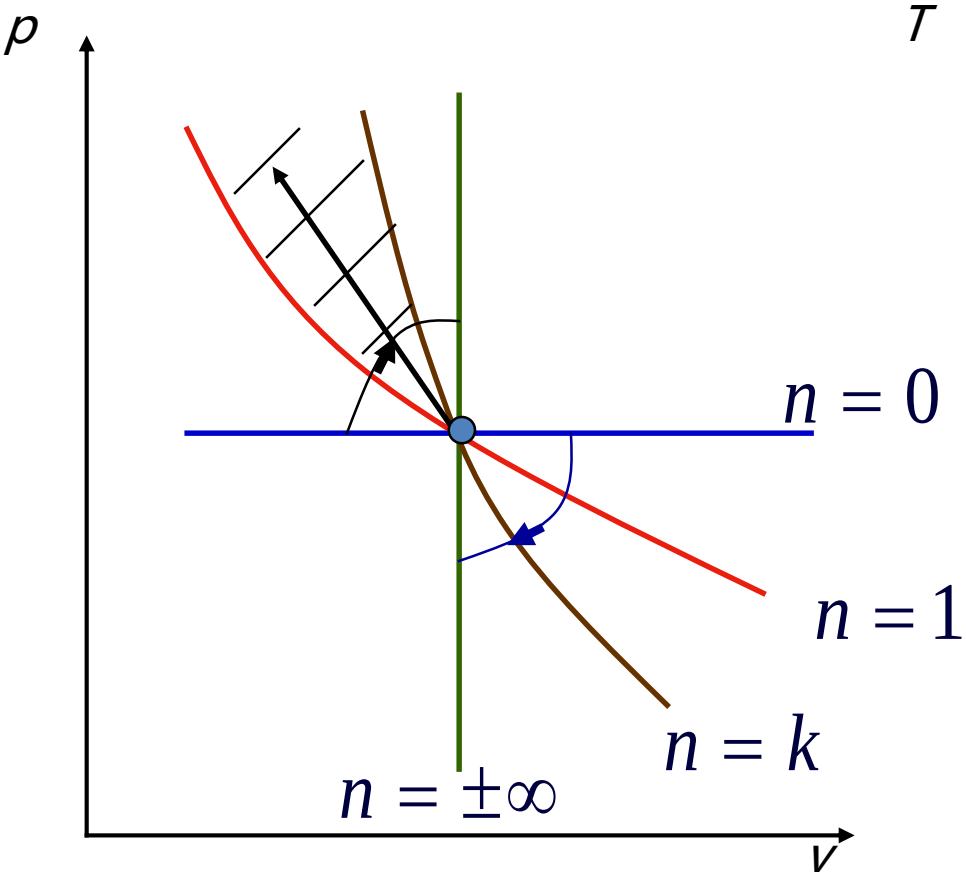
等熵线是**热量**的分界线

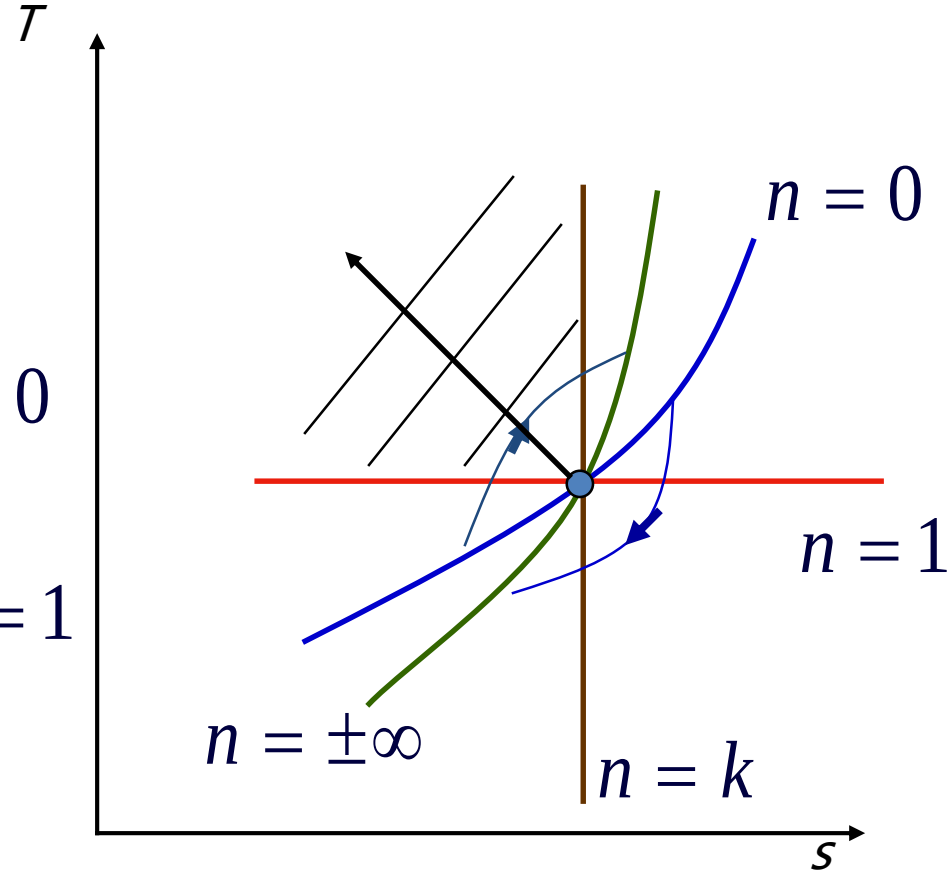
等压线是**技术功**的分界线

等容线是**体积功**的分界线

p-v, T-s 图练习（1）

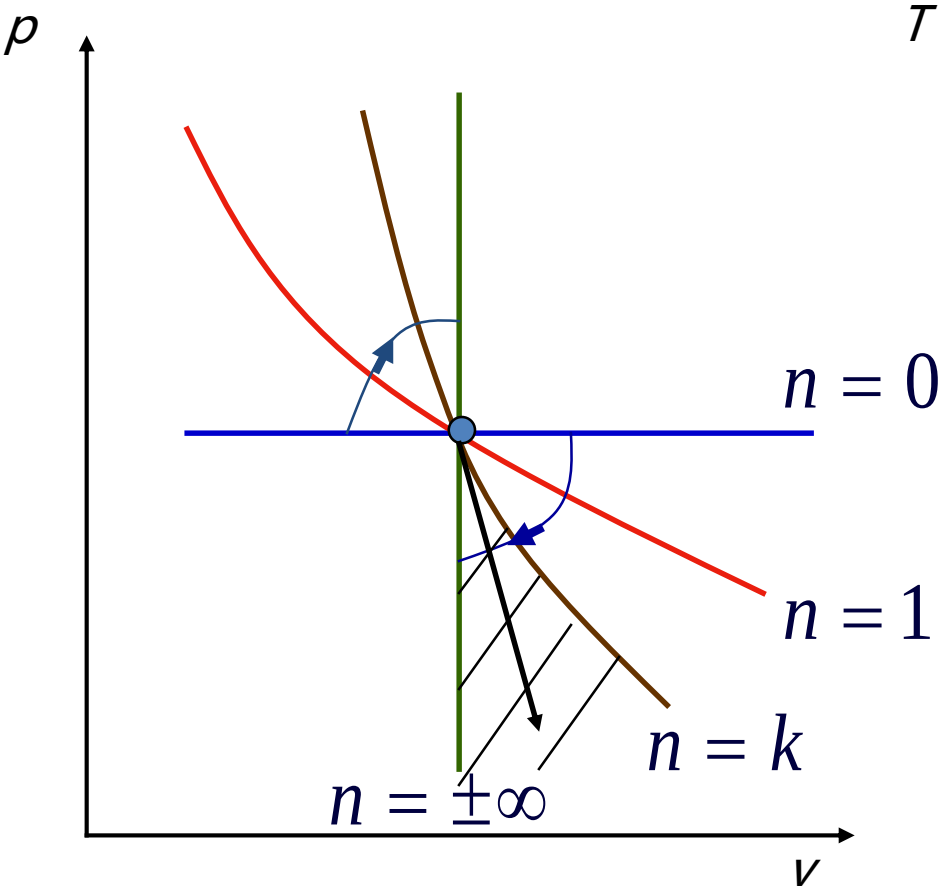
压缩、升温、放热的过程，终态在哪个区域?

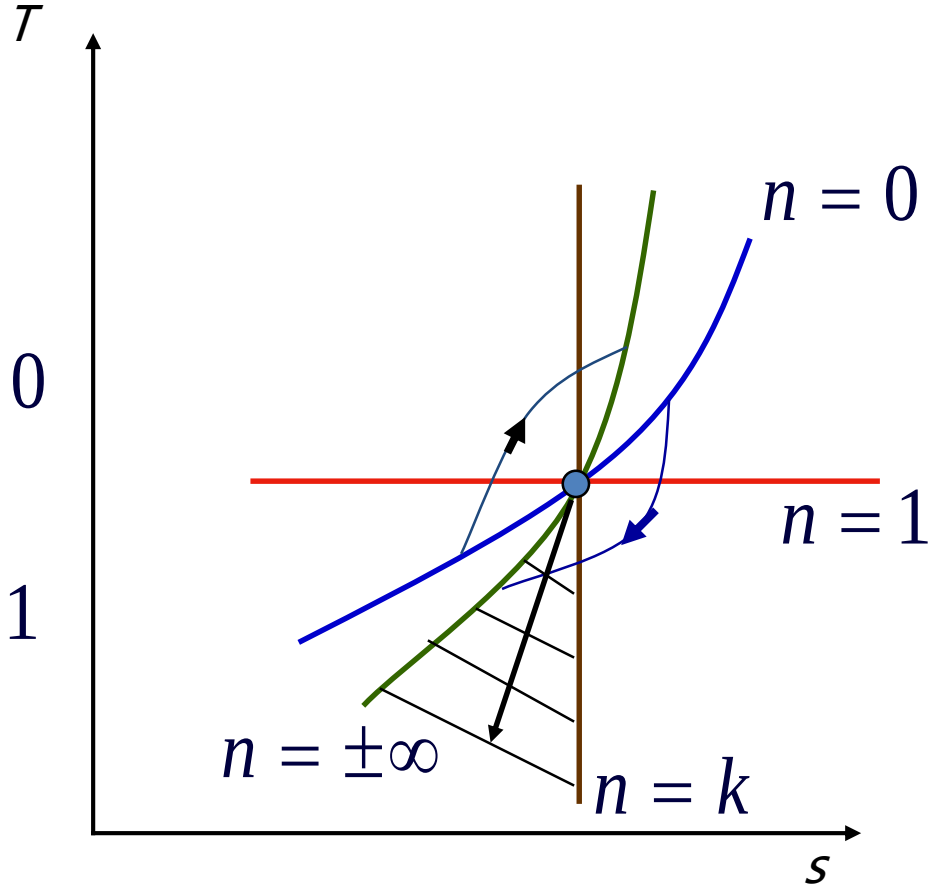




p-v, T-s 图练习（2）

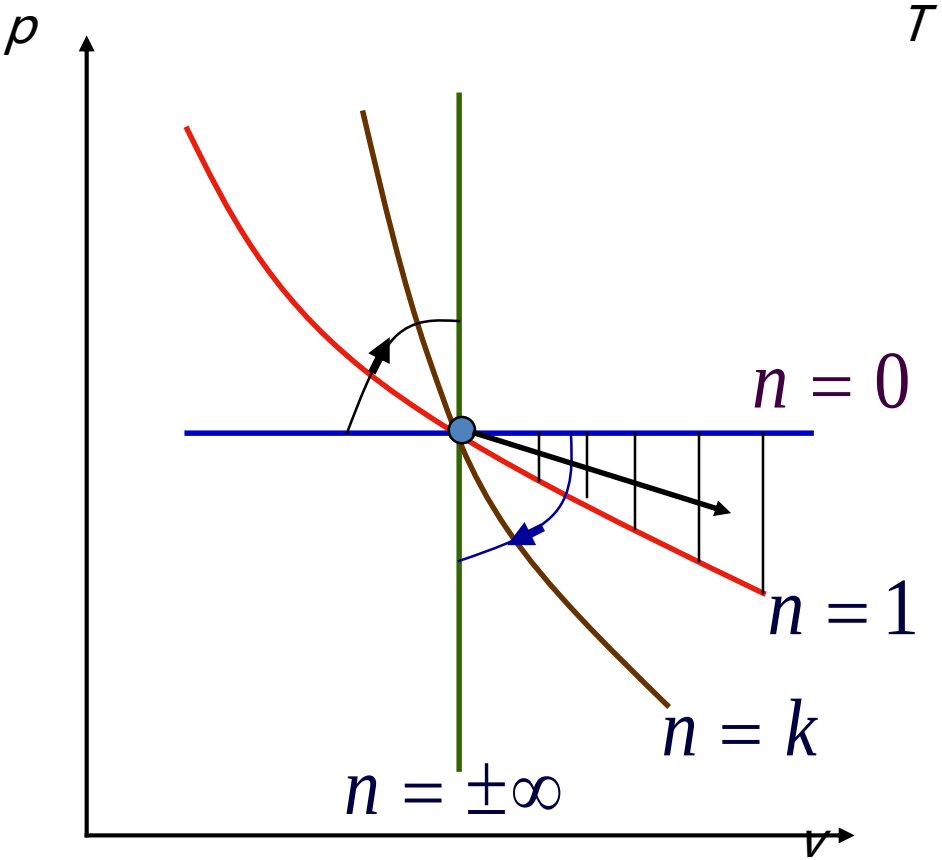
膨胀、降温、放热的过程，终态在哪个区域?

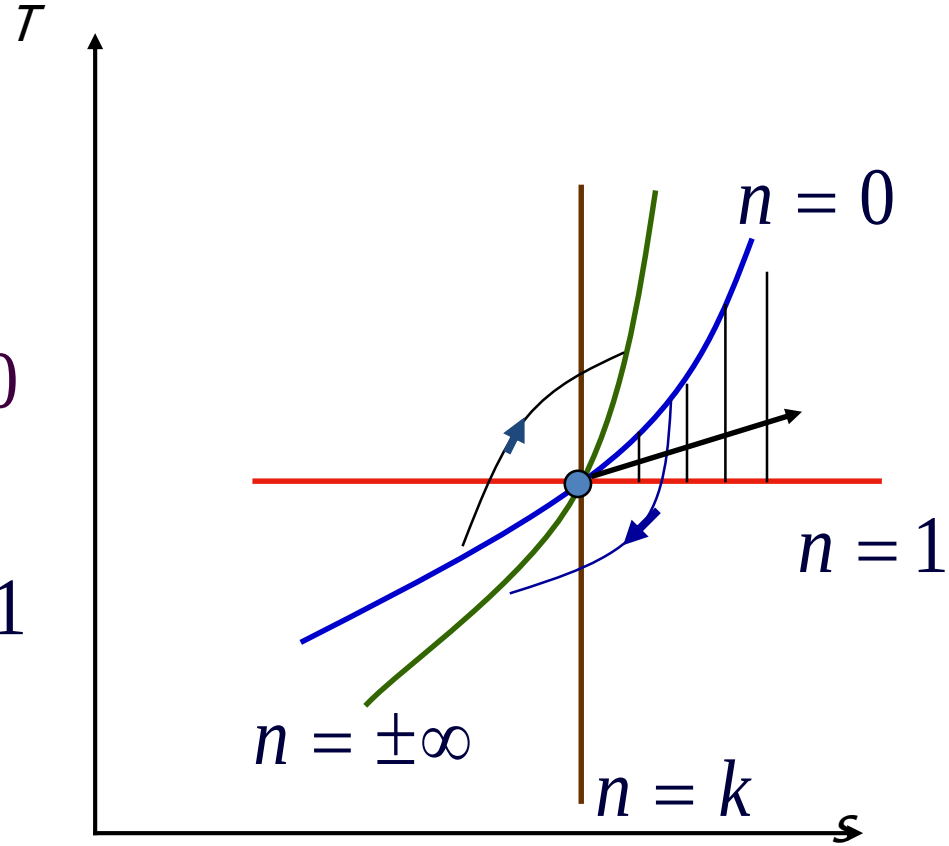




p-v, T-s 图练习（3）

膨胀、升温、吸热的过程，终态在哪个区域?





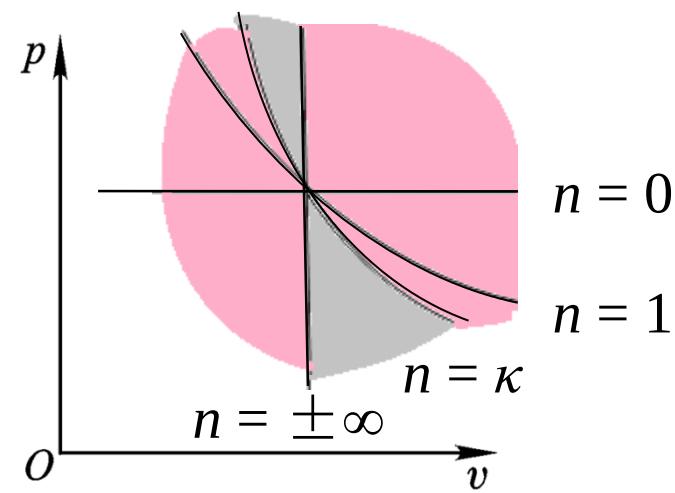
七、多变过程的能量关系 w / q

$$w=\frac{R_g}{n-1}\left(T_1-T_2\right)=\frac{\kappa-1}{n-1}c_V\left(T_1-T_2\right)$$

$$q=\frac{n-\kappa}{n-1}c_V\left(T_2-T_1\right)$$

$$\frac{w}{q}=\frac{\kappa-1}{\kappa-n}$$

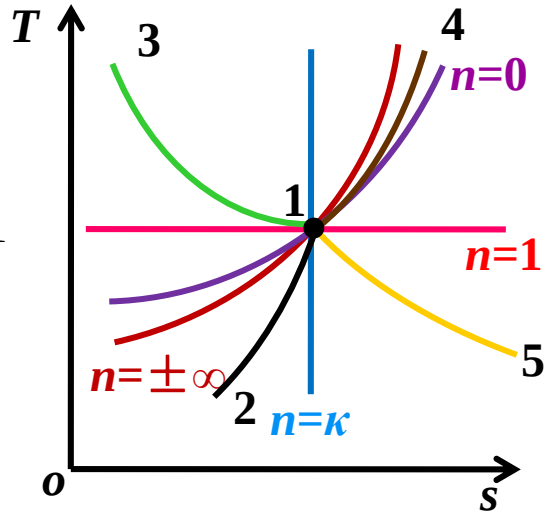
$$\Rightarrow\begin{cases} n<\kappa \quad \dfrac{\kappa-1}{\kappa-n}>0 \quad \dfrac{w}{q}>0 \\ \text{膨胀，吸热；压缩, 放热} \\ n>\kappa \quad \dfrac{\kappa-1}{\kappa-n}<0 \quad \dfrac{w}{q}<0 \\ \text{膨胀，放热；压缩, 吸热} \end{cases}$$



例题1：

某理想气体经历4个过程，1-2、1-3、1-4、1-5，如T-s图。

（1）求各过程p-v图；（2）指出过程加热或放热，膨胀或压缩。



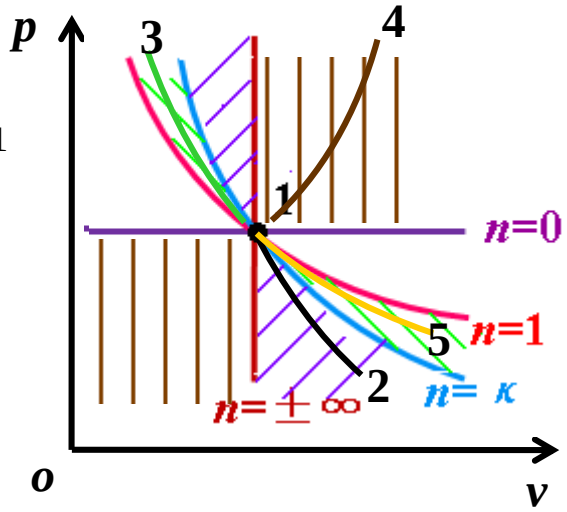
解

1-2 $\kappa < n_{1-2} < \infty \quad T_2 < T_1 \quad s_2 < s_1 \quad v_2 > v_1$

边膨胀、边放热

1-3 $1 < n_{1-3} < \kappa \quad T_3 > T_1 \quad s_3 < s_1 \quad v_3 < v_1$

边压缩、边放热



1-4 $0 > n_{1-4} > -\infty \quad T_4 > T_1 \quad s_4 > s_1 \quad v_4 > v_1$

边膨胀、边吸热

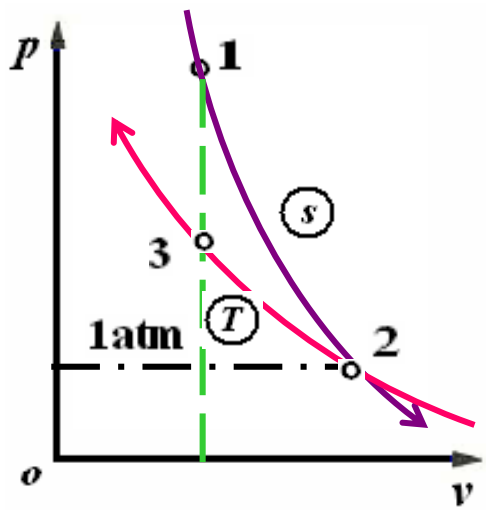
1-5 $1 < n_{1-5} < \kappa \quad T_5 < T_1 \quad s_5 > s_1 \quad v_5 > v_1$

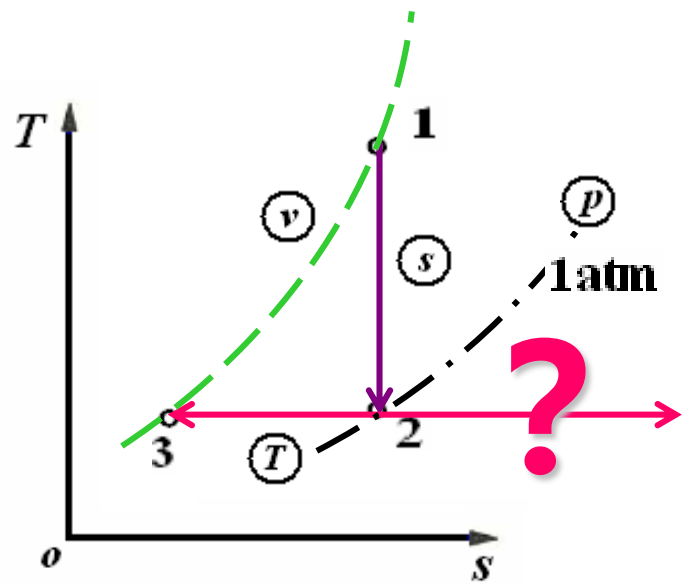
边膨胀、边吸热

例题2：

0.5 kmol某种单原子理想气体，由25 ℃，2m^3 可逆绝热膨胀到1 atm，然后在此状态的温度下定温可逆压缩回到2 m^3。1）画出各过程的*p*-*v*图及*T*-*s*图；2）计算整个过程的*Q*，*W*，Δ*U*，Δ*H* 及Δ*S*。

解　1）*p*−*v*图及*T*−*s*图





2）先求各状态参数

$$p_1 = \frac{nRT_1}{V_1} = \frac{500\ \text{mol} \times 8.314\ 5\ \text{J/(mol} \cdot \text{K)} \times (273+25)\ \text{K}}{2\ \text{m}^3}$$

$$= 6.193\ 93 \times 10^5\ \text{Pa} = 6.11\ \text{atm}$$

$$T_2 = T_1 \left(\frac{p_2}{p_1} \right)^{\frac{\kappa-1}{\kappa}} = 298\text{K} \times \left(\frac{1\text{MPa}}{6.11\text{MPa}} \right)^{\frac{1.67-1}{1.67}} = 144.26\text{K}$$

$$V_2 = V_1 \left(\frac{p_1}{p_2} \right)^{\frac{1}{\kappa}} = 2\text{m}^3 \times \left(\frac{6.11\text{MPa}}{1\text{MPa}} \right)^{\frac{1}{1.67}} = 5.906\text{m}^3$$

$$T_3 = T_2 = 144.26\ \text{K} \qquad V_3 = V_1 = 2\ \text{m}^3$$

$$p_3 = \frac{nRT_3}{V_3} = \frac{500\ \text{mol} \times 8.314\ 5\ \text{J/(mol} \cdot \text{K)} \times 144.26\ \text{K}}{2\ \text{m}^3} = 299\ 855\ \text{Pa}$$

理想气体等温过程： 热量 = 膨胀功 = 技术功

$$Q = Q_{1-2} + Q_{2-3} = Q_{2-3} = W_{T\,2-3} = nRT\ln\frac{V_3}{V_2}$$

$$= 500\ \text{mol} \times 8.314\ 5\ \text{J/(mol}\cdot\text{K)} \times 144.26\ \text{K} \times \ln\frac{2\ \text{m}^3}{5.906\ \text{m}^3}$$

$$= -649.4\ \text{kJ}$$ **放热**

$$W = W_{1-2} + W_{2-3} = \left(U_1 - U_2\right) + W_{T\,2-3} = nC_{V\text{m}}\left(T_1 - T_2\right) + nRT\ln\frac{V_3}{V_2}$$

$$= 500\ \text{mol} \times 3 \times 4.186\ 8\ \text{J/(mol}\cdot\text{K)} \times \left(298 - 144.26\right)\ \text{K} + \left(-649.4\right)\ \text{kJ}$$

$$= 316.12\ \text{kJ}$$

$$\Delta U_{1-3} = Q_{1-3} - W_{1-3} = -649.4\text{kJ} - 316.62\text{kJ} = -965.52\text{kJ}$$

$$\Delta H_{1-3} = nC_{p,\text{m}}\left(T_3 - T_1\right) = \Delta U + \Delta\left(pV\right)$$

$$= -965.52\text{kJ} + \left(299.855 - 619.393\right)\text{kPa} \times 2\text{m}^3 = 1604.6\text{kJ}$$

$$\Delta S_{1-3} = n\left[C_{V,\text{m}} \ln\frac{T_3}{T_1} + R\ln\frac{V_3}{V_1}\right]$$

$$= 500\text{mol} \times 3 \times 4.1868\text{J/(mol}\cdot\text{K)} \times \ln\frac{144.26\text{K}}{298\text{K}} = -4.56\text{kJ/K}$$

例题3：

封闭气缸中气体初态p_1 = 8MPa，t_1 =1 300℃，经可逆多变膨胀过程变化到终态 p_2 = 0.4 MPa，t_2 = 400℃。已知该气体的气体常数R_g = 0.287 kJ/(kg·K)，试判断气体在该过程中各是放热还是吸热的？气体比热容为常数，c_V = 0.716 kJ/(kg·K)。

解　对初，终态用理想气体状态方程式

$$v_1 = \frac{R_g T_1}{p_1} = \frac{287\ \text{J/(kg}\cdot\text{K)} \times (1\,300 + 273)\ \text{K}}{8\times10^6\,\text{Pa}} = 0.056\ 43\ \text{m}^3/\text{kg}$$

$$v_2 = \frac{R_g T_2}{p_2} = \frac{287\ \text{J/(kg}\cdot\text{K)} \times (400 + 273)\ \text{K}}{0.4\times10^6\ \text{Pa}} = 0.482\ 88\ \text{m}^3/\text{kg}$$

1到2是可逆多变过程，多变指数

$$n=\frac{\ln(p_1/p_2)}{\ln(v_2/v_1)}=\frac{\ln(8\times10^6\ \text{Pa}/0.4\times10^6\ \text{Pa})}{\ln(0.482\ 88\ \text{m}^3/\text{kg}/0.056\ 43\ \text{m}^3/\text{kg})}=1.395$$

多变过程膨胀功和热量

$$w=\frac{R_\text{g}}{n-1}\left(T_1-T_2\right)=\frac{287\ \text{J/(kg}\cdot\text{K)}}{1.395-1}\left(1\ 573-673\right)\ \text{K}=653.92\ \text{kJ/kg}$$

$$q=\Delta u+w=c_V\left(T_2-T_1\right)+w$$

$$=0.716\ \text{kJ/(kg}\cdot\text{K)}\times\left(673-1573\right)\ \text{K}+653.92\ \text{kJ/kg}=9.52\ \text{kJ/kg}>0$$

吸热过程

4–6 水蒸气的基本过程

一、基本公式

基本热力过程：定容、定压、定温、定熵

功、热量的计算式：

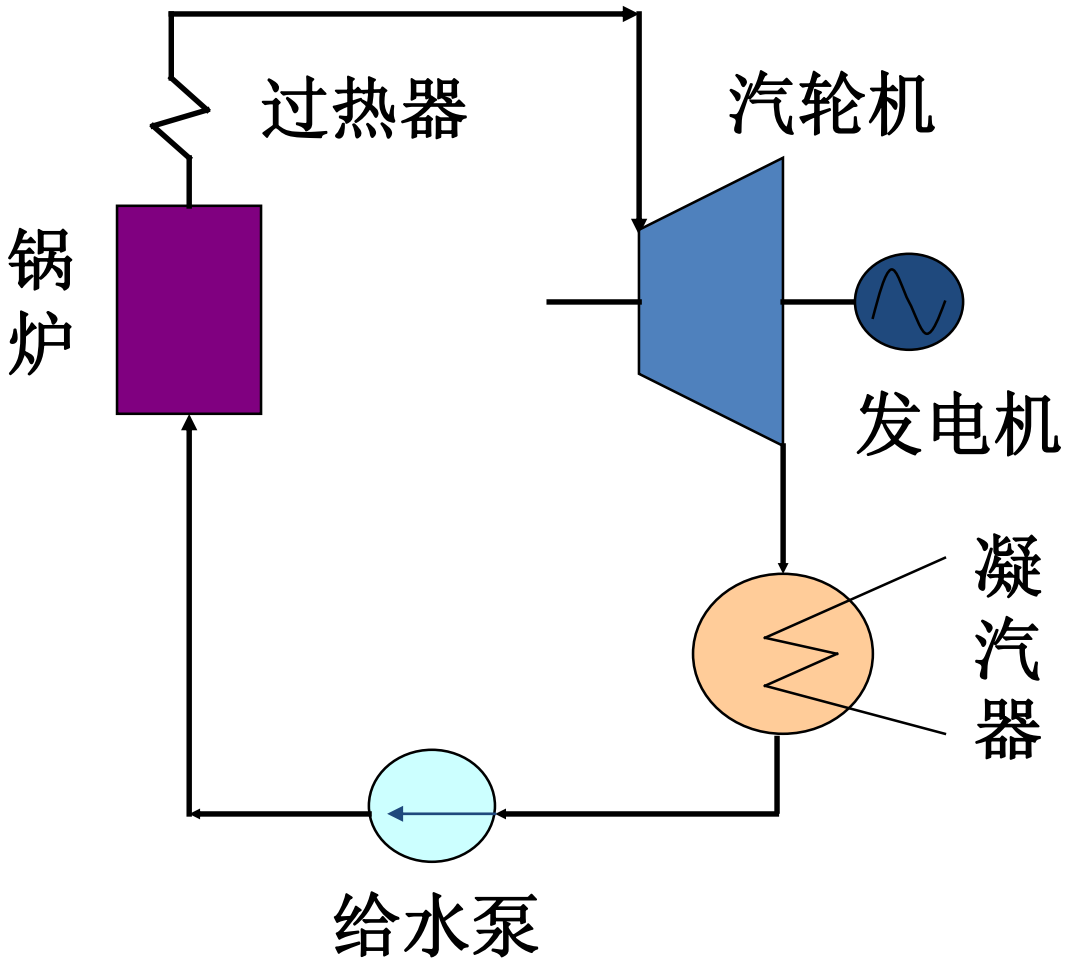
$$q = \Delta u + w \qquad q = \Delta h + w_{\mathrm{t}}$$

$$q_V = u_2 - u_1 \qquad q_p = h_2 - h_1$$

$$w = \int_1^2 p\mathrm{d}v \qquad w_{\mathrm{t}} = -\int_1^2 v\mathrm{d}p \qquad q = \int_1^2 T\mathrm{d}s$$

锅炉、过热器、凝汽器：定压过程

汽轮机、给水泵：定熵过程

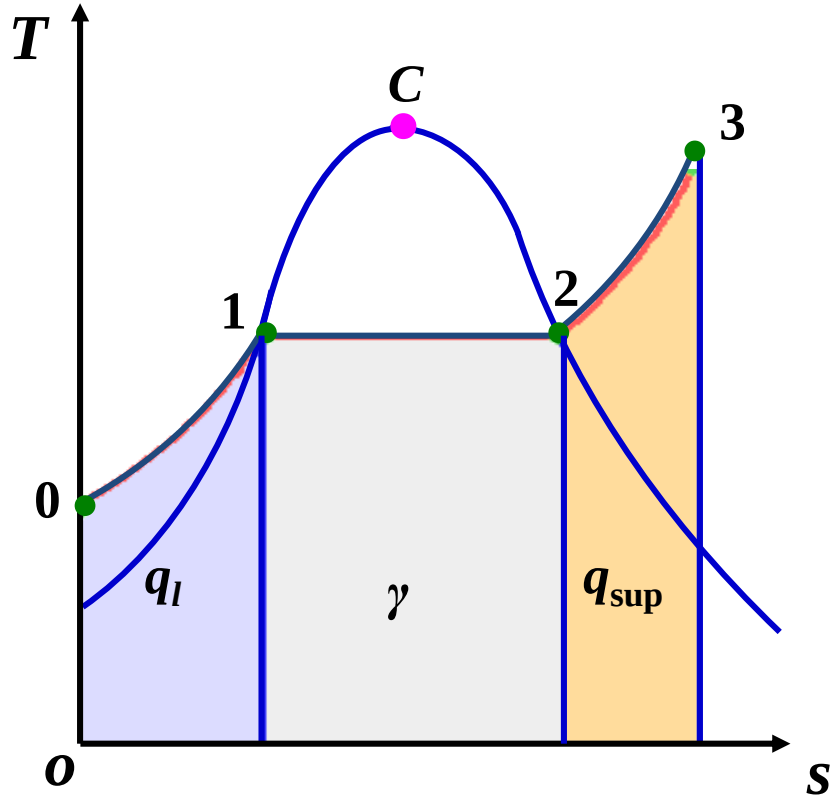


二、定压过程

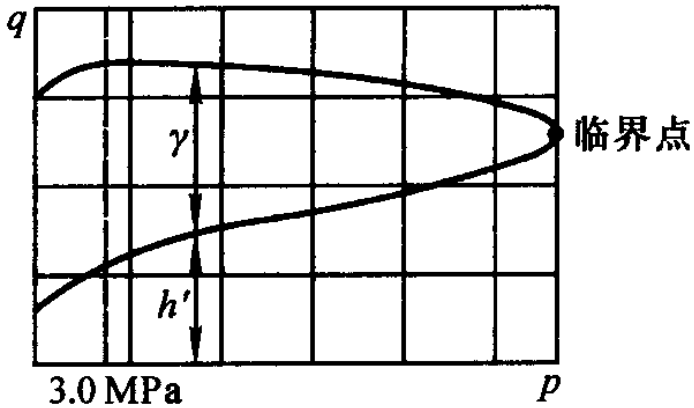
$$q_p = h_{末} - h_{初}$$

$$\begin{cases} q_l = h_1 - h_0 \\ \gamma = h_2 - h_1 \\ q_{sup} = h_3 - h_2 \end{cases}$$

$$q_p = h_3 - h_0$$



汽化潜热



p/MPa	0.001	0.01	0.1	1	10	22.12
r/(kJ/kg)	2 485.0	2 392.9	2 257.9	2 013.6	1 319.7	0

三、定熵过程

$$w_t = h_1 - h_2$$

$$w = u_1 - u_2 = (h_1 - p_1 v_1) - (h_2 - p_2 v_2)$$

注意：水蒸气

$$pv^{\kappa} = \text{const.}$$

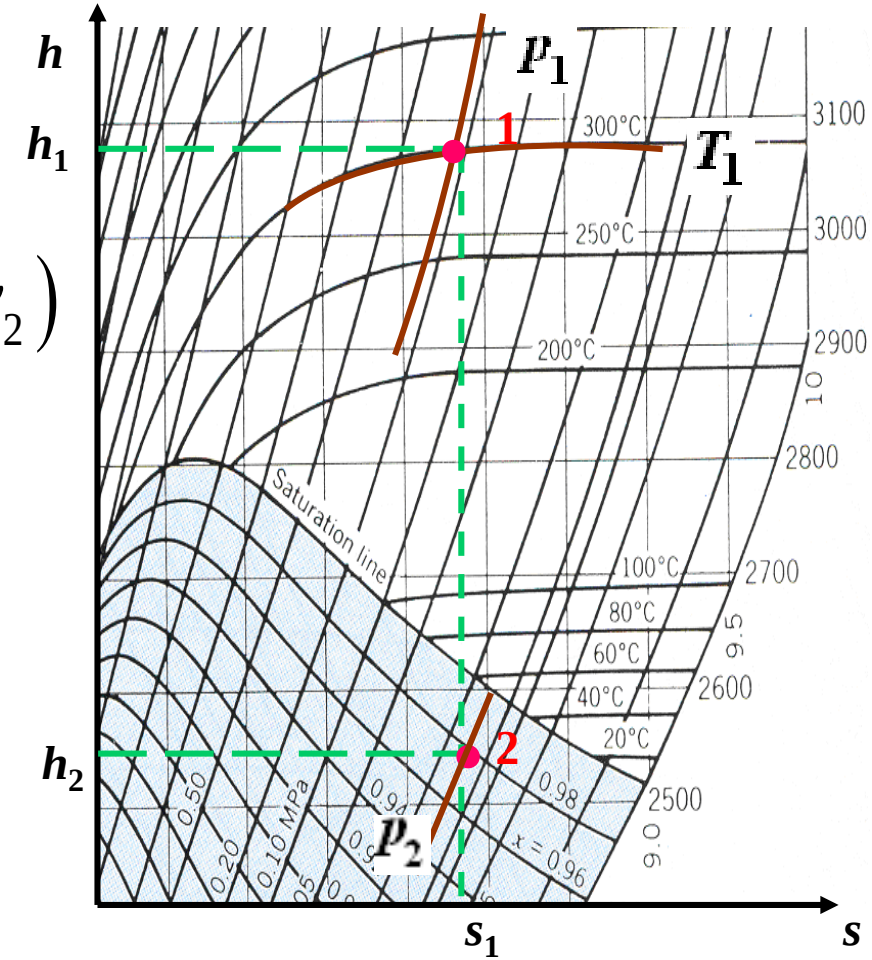
其中 $\kappa \neq \dfrac{c_p}{c_v}$ **为经验数字**

$$\left\{\begin{array}{ll} \text{过热蒸汽} & \kappa = 1.3 \\ \text{饱和蒸汽} & \kappa = 1.135 \\ \text{湿蒸汽} & \kappa = 1.035 + 0.1x \end{array}\right\}$$

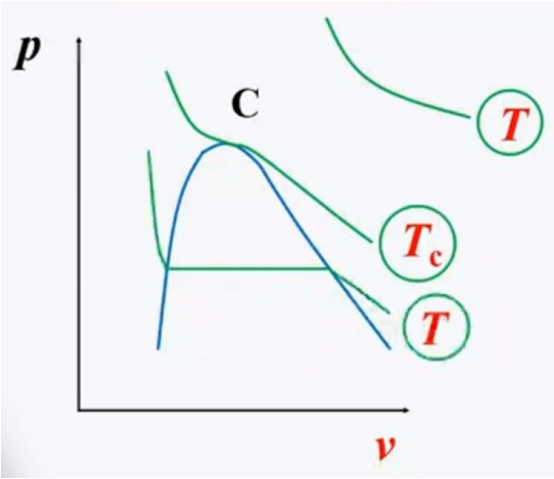
或

$$\kappa = \frac{\ln(p_1 / p_2)}{\ln(v_2 / v_1)}$$

水蒸气 $T_1 v_1^{\kappa-1} \overset{?}{=} T_2 v_2^{\kappa-1}$

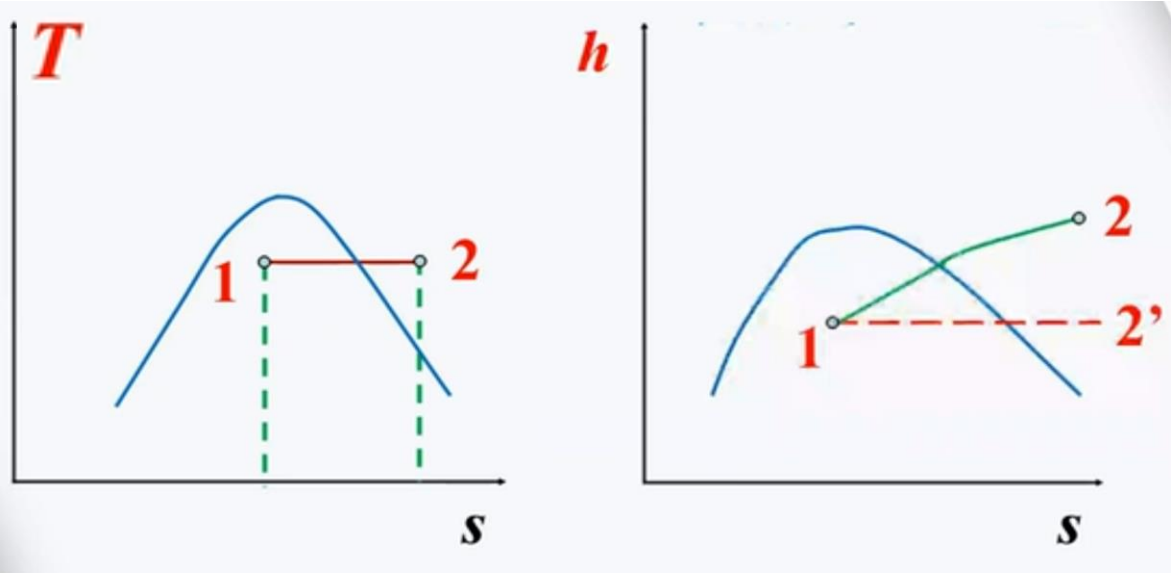


四、定温过程



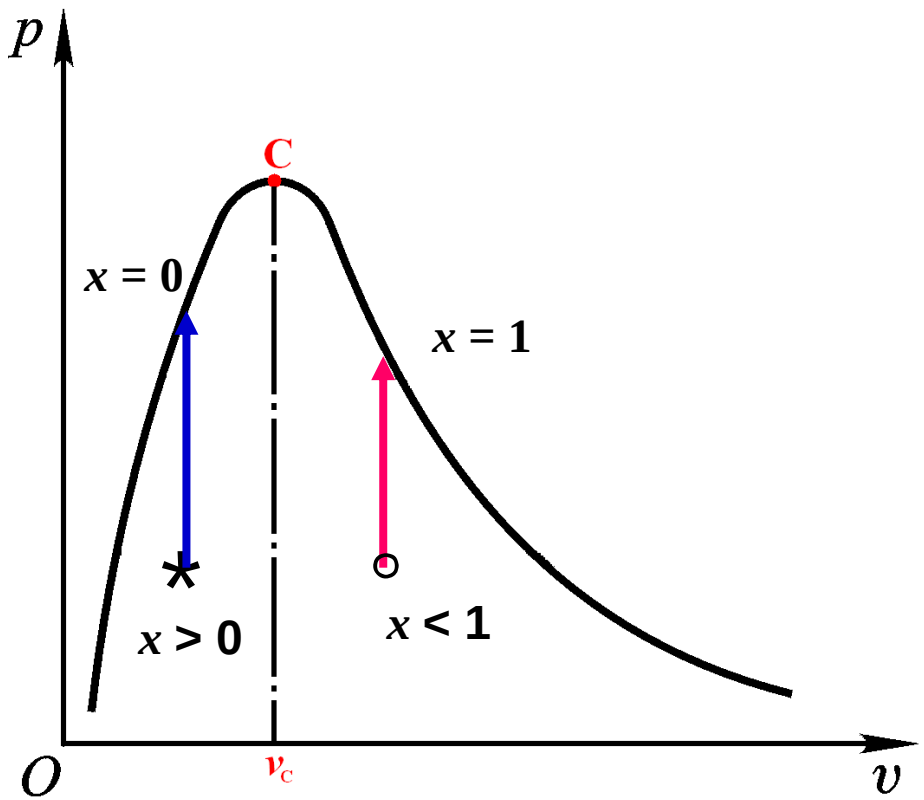
离饱和状态越远，
越接近理想气体

可逆过程： $q=\int T\mathrm{d}s=T\Delta s$ $w_t=q-\Delta h$



五、定体积过程

$$q_V = \Delta u = \Delta h - \Delta(pv)$$



思考：

定容加热时x如何变化?

$v > v_c$时，定容加热x上升

$v < v_c$时，定容加热 x下降

第4章小结

1. 过程方程、多变指数；
2. 多变过程的p-v图和T-s图；
3. 热力过程u、h和s的变化量及过程功和热量；

第4章作业

5-3，5-5，5-10，5-12，5-16

注意：只收手写作业！不收打印作业或电子版作业！